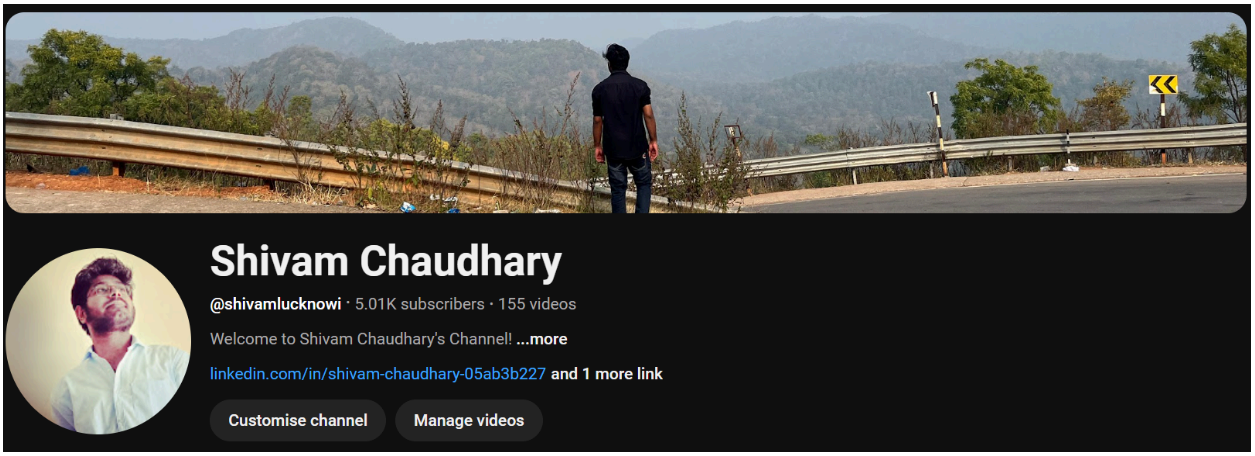


Introduction

Welcome to the ultimate resource for acing technical interviews! This sheet is a comprehensive compilation of the **most frequently asked questions** across **product-based companies, service-based companies, and startups**. It covers essential topics like **DBMS, Operating Systems (OS), Object-Oriented Programming (OOPs), and Computer Networks**—core subjects that are vital for any technical role.

Whether you're preparing for your first interview or brushing up for an advanced role, this sheet ensures you're equipped with the right knowledge to tackle the toughest questions confidently.

A huge shoutout to **Javatpoint** and **GeeksforGeeks** for their incredible content that served as a reference and inspiration. Their dedication to creating high-quality educational material has been instrumental in shaping this guide.

For more insightful content and in-depth explanations, check out my YouTube channel:
https://www.youtube.com/@shivamlucknowi

Let's get started and take a step closer to cracking your dream job! 🚀

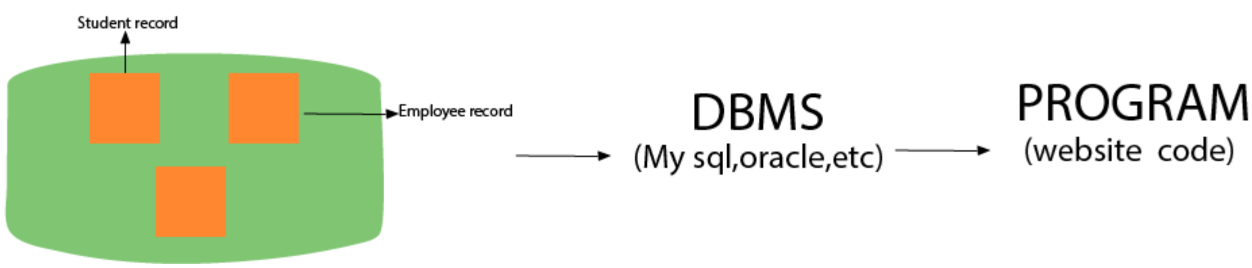
DBMS Interview Questions

A list of top frequently asked DBMS interview questions and answers are given below.

1) What is DBMS?

DBMS is a collection of programs that facilitates users to create and maintain a database. In other words, DBMS provides us an interface or tool for performing different operations such as the creation of a database, inserting data into it, deleting data from it, updating the data, etc. DBMS is a software in which data is stored in a more secure way as compared to the file-based system. Using DBMS, we can overcome many problems such as- data redundancy, data inconsistency, easy access, more organized and understandable, and so on. There is the name of some popular Database Management System- MySQL, Oracle, SQL Server, Amazon simple DB (Cloud-based), etc.

Working of DBMS is defined in the figure below.



2) What is a database?

A Database is a logical, consistent and organized collection of data that it can easily be accessed, managed and updated. Databases, also known as electronic databases are structured to provide the facility of creation, insertion, updating of the data efficiently and are stored in the form of a file or set of files, on the magnetic disk, tapes and another sort of secondary devices. Database mostly consists of the objects (tables), and tables include of the records and fields. Fields are the basic units of data storage, which contain the information about a particular aspect or attribute of the entity described by the database. DBMS is used for extraction of data from the database in the form of the queries.

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3) What is a database system?

The collection of database and DBMS software together is known as a database system. Through the database system, we can perform many activities such as-

The data can be stored in the database with ease, and there are no issues of data redundancy and data inconsistency.

The data will be extracted from the database using DBMS software whenever required. So, the combination of database and DBMS software enables one to store, retrieve and access data with considerate accuracy and security.

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4) What are the advantages of DBMS?

- Redundancy control
- Restriction for unauthorized access
- Provides multiple user interfaces
- Provides backup and recovery
- Enforces integrity constraints
- Ensure data consistency
- Easy accessibility
- Easy data extraction and data processing due to the use of queries

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5) What is a checkpoint in DBMS?

The Checkpoint is a type of mechanism where all the previous logs are removed from the system and permanently stored in the storage disk.

There are two ways which can help the DBMS in recovering and maintaining the ACID properties, and they are- maintaining the log of each transaction and maintaining shadow pages. So, when it comes to log based recovery system, checkpoints come into existence. Checkpoints are those points to which the database engine can recover after a crash as a specified minimal point from where the transaction log record can be used to recover all the committed data up to the point of the crash.

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6) When does checkpoint occur in DBMS?

A checkpoint is like a snapshot of the DBMS state. Using checkpoints, the DBMS can reduce the amount of work to be done during a restart in the event of subsequent crashes. Checkpoints are used for the recovery of the database after the system crash. Checkpoints are used in the log-based recovery system. When due to a system crash we need to restart the system then at that point we use checkpoints. So that, we don't have to perform the transactions from the very starting.

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7) What do you mean by transparent DBMS?

The transparent DBMS is a type of DBMS which keeps its physical structure hidden from users. Physical structure or physical storage structure implies to the memory manager of the DBMS, and it describes how the data stored on disk.

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8) What are the unary operations in Relational Algebra?

PROJECTION and SELECTION are the unary operations in relational algebra. Unary operations are those operations which use single operands. Unary operations are SELECTION, PROJECTION, and RENAME.

As in SELECTION relational operators are used for example - =,<=,>=, etc.

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9) What is RDBMS?

RDBMS stands for Relational Database Management Systems. It is used to maintain the data records and indices in tables. RDBMS is the form of DBMS which uses the structure to identify and access data concerning the other piece of data in the database. RDBMS is the system that enables you to perform different operations such as- update, insert, delete, manipulate and administer a relational database with minimal difficulties. Most of the time RDBMS use SQL language because it is easily understandable and is used for often.

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10) How many types of database languages are?

There are four types of database languages:

- Data Definition Language (DDL) e.g., CREATE, ALTER, DROP, TRUNCATE, RENAME, etc. All these commands are used for updating the data that?s why they are known as Data Definition Language.
- Data Manipulation Language (DML) e.g., SELECT, UPDATE, INSERT, DELETE, etc. These commands are used for the manipulation of already updated data that's why they are the part of Data Manipulation Language.
- DATA Control Language (DCL) e.g., GRANT and REVOKE. These commands are used for giving and removing the user access on the database. So, they are the part of Data Control Language.
- Transaction Control Language (TCL) e.g., COMMIT, ROLLBACK, and SAVEPOINT. These are the commands used for managing transactions in the database. TCL is used for managing the changes made by DML.

Database language implies the queries that are used for the update, modify and manipulate the data.

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11) What do you understand by Data Model?

The Data model is specified as a collection of conceptual tools for describing data, data relationships, data semantics and constraints. These models are used to describe the relationship between the entities and their attributes.

There is the number of data models:

- Hierarchical data model
- network model
- relational model
- Entity-Relationship model and so on.

12) Define a Relation Schema and a Relation.

A Relation Schema is specified as a set of attributes. It is also known as table schema. It defines what the name of the table is. Relation schema is known as the blueprint with the help of which we can explain that how the data is organized into tables. This blueprint contains no data.

A relation is specified as a set of tuples. A relation is the set of related attributes with identifying key attributes

See this example:

Let r be the relation which contains set tuples (t1, t2, t3, ..., tn). Each tuple is an ordered list of n-values t=(v1,v2, ..., vn).

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13) What is a degree of Relation?

The degree of relation is a number of attribute of its relation schema. A degree of relation is also known as Cardinality it is defined as the number of occurrence of one entity which is connected to the number of occurrence of other entity. There are three degree of relation they are one-to-one(1:1), one-to-many(1:M), many-to-one(M:M).

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14) What is the Relationship?

The Relationship is defined as an association among two or more entities. There are three type of relationships in DBMS-

One-To-One: Here one record of any object can be related to one record of another object.

One-To-Many (many-to-one): Here one record of any object can be related to many records of other object and vice versa.

Many-to-many: Here more than one records of an object can be related to n number of records of another object.

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15) What are the disadvantages of file processing systems?

- Inconsistent
- Not secure
- Data redundancy
- Difficult in accessing data
- Data isolation
- Data integrity
- Concurrent access is not possible
- Limited data sharing
- Atomicity problem

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16) What is data abstraction in DBMS?

Data abstraction in DBMS is a process of hiding irrelevant details from users. Because database systems are made of complex data structures so, it makes accessible the user interaction with the database.

For example: We know that most of the users prefer those systems which have a simple GUI that means no complex processing. So, to keep the user tuned and for making the access to the data easy, it is necessary to do data abstraction. In addition to it, data abstraction divides the system in different layers to make the work specified and well defined.

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17) What are the three levels of data abstraction?

Following are three levels of data abstraction:

Physical level: It is the lowest level of abstraction. It describes how data are stored.

Logical level: It is the next higher level of abstraction. It describes what data are stored in the database and what the relationship among those data is.

View level: It is the highest level of data abstraction. It describes only part of the entire database.

For example- User interacts with the system using the GUI and fill the required details, but the user doesn't have any idea how the data is being used. So, the abstraction level is entirely high in VIEW LEVEL.

Then, the next level is for PROGRAMMERS as in this level the fields and records are visible and the programmers have the knowledge of this layer. So, the level of abstraction here is a little low in VIEW LEVEL.

And lastly, physical level in which storage blocks are described.

---

18) What is DDL (Data Definition Language)?

Data Definition Language (DDL) is a standard for commands which defines the different structures in a database. Most commonly DDL statements are CREATE, ALTER, and DROP. These commands are used for updating data into the database.

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19) What is DML (Data Manipulation Language)?

DData Manipulation Language (DML) is a language that enables the user to access or manipulate data as organized by the appropriate data model. For example- SELECT, UPDATE, INSERT, DELETE.

There is two type of DML:

Procedural DML or Low level DML: It requires a user to specify what data are needed and how to get those data.

Non-Procedural DML or High level DML:It requires a user to specify what data are needed without specifying how to get those data.

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20) Explain the functionality of DML Compiler.

The DML Compiler translates DML statements in a query language that the query evaluation engine can understand. DML Compiler is required because the DML is the family of syntax element which is very similar to the other programming language which requires compilation. So, it is essential to compile the code in the language which query evaluation engine can understand and then work on those queries with proper output.

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21) What is Relational Algebra?

Relational Algebra is a Procedural Query Language which contains a set of operations that take one or two relations as input and produce a new relationship. Relational algebra is the basic set of operations for the relational model. The decisive point of relational algebra is that it is similar to the algebra which operates on the number.

There are few fundamental operations of relational algebra:

- select
- project
- set difference
- union
- rename,etc.

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22) What is Relational Calculus?

Relational Calculus is a Non-procedural Query Language which uses mathematical predicate calculus instead of algebra. Relational calculus doesn't work on mathematics fundamentals such as algebra, differential, integration, etc. That's why it is also known as predicate calculus.

There is two type of relational calculus:

- Tuple relational calculus
- Domain relational calculus

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23) What do you understand by query optimization?

The term query optimization specifies an efficient execution plan for evaluating a query that has the least estimated cost. The concept of query optimization came into the frame when there were a number of methods, and algorithms existed for the same task then the question arose that which one is more efficient and the process of determining the efficient way is known as query optimization.

There are many benefits of query optimization:

- It reduces the time and space complexity.
- More queries can be performed as due to optimization every query comparatively takes less time.
- User satisfaction as it will provide output fast

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24) What do you mean by durability in DBMS?

Once the DBMS informs the user that a transaction has completed successfully, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called durability. Durability ensures that once the transaction is committed into the database, it will be stored in the non-volatile memory and after that system failure cannot affect that data anymore.

---

25) What is normalization?

Normalization is a process of analysing the given relation schemas according to their functional dependencies. It is used to minimize redundancy and also used to minimize insertion, deletion and update distractions. Normalization is considered as an essential process as it is used to avoid data redundancy, insertion anomaly, updation anomaly, deletion anomaly.

There most commonly used normal forms are:

- First Normal Form(1NF)
- Second Normal Form(2NF)
- Third Normal Form(3NF)
- Boyce & Codd Normal Form(BCNF)

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26) What is Denormalization?

Denormalization is the process of boosting up database performance and adding of redundant data which helps to get rid of complex data. Denormalization is a part of database optimization technique. This process is used to avoid the use of complex and costly joins. Denormalization doesn't refer to the thought of not to normalize instead of that denormalization takes place after normalization. In this process, firstly the redundancy of the data will be removed using normalization process than through denormalization process we will add redundant data as per the requirement so that we can easily avoid the costly joins.

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27) What is functional Dependency?

Functional Dependency is the starting point of normalization. It exists when a relation between two attributes allow you to determine the corresponding attribute's value uniquely. The functional dependency is also known as database dependency and defines as the relationship which occurs when one attribute in a relation uniquely determines another attribute. It is written as A->B which means B is functionally dependent on A.

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28) What is the E-R model?

E-R model is a short name for the Entity-Relationship model. This model is based on the real world. It contains necessary objects (known as entities) and the relationship among these objects. Here the primary objects are the entity, attribute of that entity, relationship set, an attribute of that relationship set can be mapped in the form of E-R diagram.

In E-R diagram, entities are represented by rectangles, relationships are represented by diamonds, attributes are the characteristics of entities and represented by ellipses, and data flow is represented through a straight line.

---

29) What is an entity?

The Entity is a set of attributes in a database. An entity can be a real-world object which physically exists in this world. All the entities have their attribute which in the real world considered as the characteristics of the object.

For example: In the employee database of a company, the employee, department, and the designation can be considered as the entities. These entities have some characteristics which will be the attributes of the corresponding entity.

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30) What is an Entity type?

An entity type is specified as a collection of entities, having the same attributes. Entity type typically corresponds to one or several related tables in the database. A characteristic or trait which defines or uniquely identifies the entity is called entity type.

For example, a student has student_id, department, and course as its characteristics.

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31) What is an Entity set?

The entity set specifies the collection of all entities of a particular entity type in the database. An entity set is known as the set of all the entities which share the same properties.

For example, a set of people, a set of students, a set of companies, etc.

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32) What is an Extension of entity type?

An extension of an entity type is specified as a collection of entities of a particular entity type that are grouped into an entity set.

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33) What is Weak Entity set?

An entity set that doesn't have sufficient attributes to form a primary key is referred to as a weak entity set. The member of a weak entity set is known as a subordinate entity. Weak entity set does not have a primary key, but we need a mean to differentiate among all those entries in the entity set that depend on one particular strong entity set.

34) What is an attribute?

An attribute refers to a database component. It is used to describe the property of an entity. An attribute can be defined as the characteristics of the entity. Entities can be uniquely identified using the attributes. Attributes represent the instances in the row of the database.

For example: If a student is an entity in the table then age will be the attribute of that student.

35) What are the integrity rules in DBMS?

Data integrity is one significant aspect while maintaining the database. So, data integrity is enforced in the database system by imposing a series of rules. Those set of integrity is known as the integrity rules.

There are two integrity rules in DBMS:

Entity Integrity : It specifies that "Primary key cannot have a NULL value."

Referential Integrity: It specifies that "Foreign Key can be either a NULL value or should be the Primary Key value of other relation

36) What do you mean by extension and intension?

Extension: The Extension is the number of tuples present in a table at any instance. It changes as the tuples are created, updated and destroyed. The actual data in the database change quite frequently. So, the data in the database at a particular moment in time is known as extension or database state or snapshot. It is time dependent.

Intension: Intension is also known as Data Schema and defined as the description of the database, which is specified during database design and is expected to remain unchanged. The Intension is a constant value that gives the name, structure of tables and the constraints laid on it.

37) What is System R? How many of its two major subsystems?

System R was designed and developed from 1974 to 1979 at IBM San Jose Research Centre. System R is the first implementation of SQL, which is the standard relational data query language, and it was also the first to demonstrate that RDBMS could provide better transaction processing performance. It is a prototype which is formed to show that it is possible to build a Relational System that can be used in a real-life environment to solve real-life problems.

Following are two major subsystems of System R:

- Research Storage
- System Relational Data System

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38) What is Data Independence?

Data independence specifies that "the application is independent of the storage structure and access strategy of data." It makes you able to modify the schema definition at one level without altering the schema definition in the next higher level.

It makes you able to modify the schema definition in one level should not affect the schema definition in the next higher level.

There are two types of Data Independence:

Physical Data Independence: Physical data is the data stored in the database. It is in the bit-format. Modification in physical level should not affect the logical level.

For example: If we want to manipulate the data inside any table that should not change the format of the table.

Logical Data Independence: Logical data in the data about the database. It basically defines the structure. Such as tables stored in the database. Modification in logical level should not affect the view level.

For example: If we need to modify the format of any table, that modification should not affect the data inside it.

NOTE: Logical Data Independence is more difficult to achieve.

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39) What are the three levels of data abstraction?

Following are three levels of data abstraction:

Physical level: It is the lowest level of abstraction. It describes how data are stored.

Logical level: It is the next higher level of abstraction. It describes what data are stored in the database and what relationship among those data.

View level: It is the highest level of data abstraction. It describes only part of the entire database.

For example- User interact with the system using the GUI and fill the required details, but the user doesn't have any idea how the data is being used. So, the abstraction level is absolutely high in VIEW LEVEL.

Then, the next level is for PROGRAMMERS as in this level the fields and records are visible and the programmer has the knowledge of this layer. So, the level of abstraction here is a little low in VIEW LEVEL.

And lastly, physical level in which storage blocks are described.

---

40) What is Join?

The Join operation is one of the most useful activities in relational algebra. It is most commonly used way to combine information from two or more relations. A Join is always performed on the basis of the same or related column. Most complex queries of SQL involve JOIN command.

There are following types of join:

- Inner joins: Inner join is of 3 categories. They are:
    - Theta join

    - Natural join
    - Equi join
- Outer joins: Outer join have three types. They are:
    - Left outer join
    - Right outer join
    - Full outer join

---

41) What is 1NF?

1NF is the First Normal Form. It is the simplest type of normalization that you can implement in a database. The primary objectives of 1NF are to:

- Every column must have atomic (single value)
- To Remove duplicate columns from the same table
- Create separate tables for each group of related data and identify each row with a unique column

---

42) What is 2NF?

2NF is the Second Normal Form. A table is said to be 2NF if it follows the following conditions:

- The table is in 1NF, i.e., firstly it is necessary that the table should follow the rules of 1NF.
- Every non-prime attribute is fully functionally dependent on the primary key, i.e., every non-key attribute should be dependent on the primary key in such a way that if any key element is deleted, then even the non_key element will still be saved in the database.

---

43) What is 3NF?

3NF stands for Third Normal Form. A database is called in 3NF if it satisfies the following conditions:

- It is in second normal form.
- There is no transitive functional dependency.
- For example: X->Z

Where:
X->Y
Y does not -> X
Y->Z so, X->Z

---

44) What is BCNF?

BCMF stands for Boyce-Codd Normal Form. It is an advanced version of 3NF, so it is also referred to as 3.5NF. BCNF is stricter than 3NF.

A table complies with BCNF if it satisfies the following conditions:

- It is in 3NF.
- For every functional dependency X->Y, X should be the super key of the table. It merely means that X cannot be a non-prime attribute if Y is a prime attribute.

---

45) Explain ACID properties

ACID properties are some basic rules, which has to be satisfied by every transaction to preserve the integrity. These properties and rules are:

ATOMICITY: Atomicity is more generally known as ?all or nothing rule.' Which implies all are considered as one unit, and they either run to completion or not executed at all.

CONSISTENCY: This property refers to the uniformity of the data. Consistency implies that the database is consistent before and after the transaction.

ISOLATION: This property states that the number of the transaction can be executed concurrently without leading to the inconsistency of the database state.

DURABILITY: This property ensures that once the transaction is committed it will be stored in the non-volatile memory and system crash can also not affect it anymore.

---

46) What is stored procedure?

A stored procedure is a group of SQL statements that have been created and stored in the database. The stored procedure increases the reusability as here the code or the procedure is stored into the system and used again and again that makes the work easy, takes less time in processing and decreases the complexity of the system. So, if you have a code which you need to use again and again then save that code and call that code whenever it is required.

---

47) What is the difference between a DELETE command and TRUNCATE command?

DELETE command: DELETE command is used to delete rows from a table based on the condition that we provide in a WHERE clause.

- DELETE command delete only those rows which are specified with the WHERE clause.
- DELETE command can be rolled back.
- DELETE command maintain a log, that's why it is slow.
- DELETE use row lock while performing DELETE function.

TRUNCATE command: TRUNCATE command is used to remove all rows (complete data) from a table. It is similar to the DELETE command with no WHERE clause.

- The TRUNCATE command removes all the rows from the table.
- The TRUNCATE command cannot be rolled back.
- The TRUNCATE command doesn't maintain a log. That's why it is fast.

- TRUNCATE use table log while performing the TRUNCATE function.

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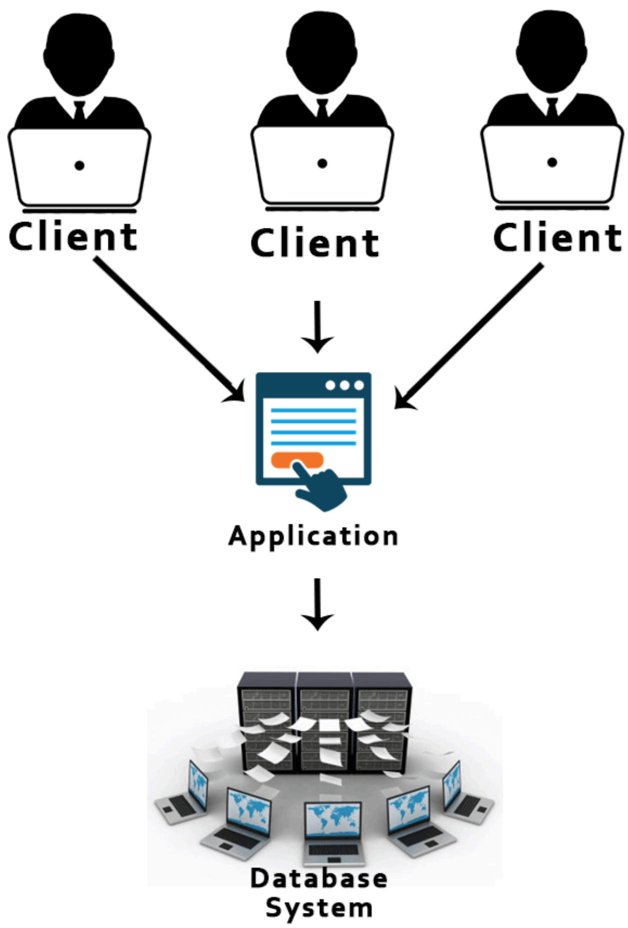
48) What is 2-Tier architecture?

The 2-Tier architecture is the same as basic client-server. In the two-tier architecture, applications on the client end can directly communicate with the database at the server side.



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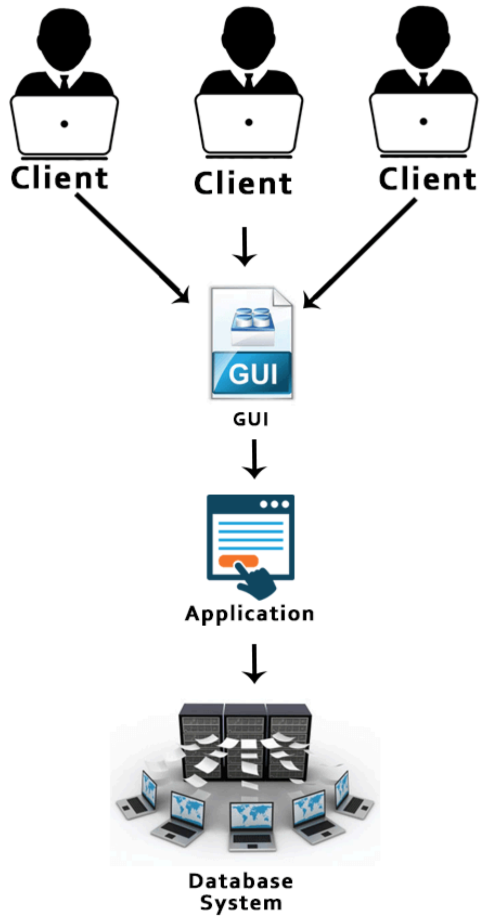
49) What is the 3-Tier architecture?

The 3-Tier architecture contains another layer between the client and server. Introduction of 3-tier architecture is for the ease of the users as it provides the GUI, which, make the system secure and much more accessible. In this architecture, the application on the client-end interacts with an application on the server which further communicates with the database system.



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50) How do you communicate with an RDBMS?

You have to use Structured Query Language (SQL) to communicate with the RDBMS. Using queries of SQL, we can give the input to the database and then after processing of the queries database will provide us the required output.

---

51) What is the difference between a shared lock and exclusive lock?

Shared lock: Shared lock is required for reading a data item. In the shared lock, many transactions may hold a lock on the same data item. When more than one transaction is allowed to read the data items then that is known as the shared lock.

Exclusive lock: When any transaction is about to perform the write operation, then the lock on the data item is an exclusive lock. Because, if we allow more than one transaction then that will lead to the inconsistency in the database.

---

52) Describe the types of keys?

There are following types of keys:

Primary key: The Primary key is an attribute in a table that can uniquely identify each record in a table. It is compulsory for every table.

Candidate key: The Candidate key is an attribute or set of an attribute which can uniquely identify a tuple. The Primary key can be selected from these attributes.

Super key: The Super key is a set of attributes which can uniquely identify a tuple. Super key is a superset of the candidate key.

Foreign key: The Foreign key is a primary key from one table, which has a relationship with another table. It acts as a cross-reference between tables.

Operating System Questions

A list of top frequently asked Operating System interview questions and answers are given below.

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1) What is an operating system?

The operating system is a software program that facilitates computer hardware to communicate and operate with the computer software. It is the most important part of a computer system without it computer is just like a box.

---

2) What is the main purpose of an operating system?

There are two main purposes of an operating system:

- It is designed to make sure that a computer system performs well by managing its computational activities.
- It provides an environment for the development and execution of programs.

---

3) What are the different operating systems?

- Batched operating systems
- Distributed operating systems
- Timesharing operating systems
- Multi-programmed operating systems
- Real-time operating systems

---

4) What is a socket?

A socket is used to make connection between two applications. Endpoints of the connection are called socket.

---

5) What is a real-time system?

Real-time system is used in the case when rigid-time requirements have been placed on the operation of a processor. It contains a well defined and fixed time constraints.

---

6) What is kernel?

Kernel is the core and most important part of a computer operating system which provides basic services for all parts of the OS.

---

7) What is monolithic kernel?

A monolithic kernel is a kernel which includes all operating system code is in single executable image.

---

8) What do you mean by a process?

An executing program is known as process. There are two types of processes:

- Operating System Processes
- User Processes

---

9) What are the different states of a process?

A list of different states of process:

- New Process
- Running Process
- Waiting Process
- Ready Process
- Terminated Process

---

10) What is the difference between micro kernel and macro kernel?

Micro kernel: micro kernel is the kernel which runs minimal performance affecting services for operating system. In micro kernel operating system all other operations are performed by processor.

Macro Kernel: Macro Kernel is a combination of micro and monolithic kernel.

---

11) What is the concept of reentrancy?

It is a very useful memory saving technique that is used for multi-programmed time sharing systems. It provides functionality that multiple users can share a single copy of program during the same period.

It has two key aspects:

- The program code cannot modify itself.
- The local data for each user process must be stored separately.

---

12) What is the difference between process and program?

A program while running or executing is known as a process.

---

13) What is the use of paging in operating system?

Paging is used to solve the external fragmentation problem in operating system. This technique ensures that the data you need is available as quickly as possible.

---

14) What is the concept of demand paging?

Demand paging specifies that if an area of memory is not currently being used, it is swapped to disk to make room for an application's need.

---

15) What is the advantage of a multiprocessor system?

As many as processors are increased, you will get the considerable increment in throughput. It is cost effective also because they can share resources. So, the overall reliability increases.

---

16) What is virtual memory?

Virtual memory is a very useful memory management technique which enables processes to execute outside of memory. This technique is especially used when an executing program cannot fit in the physical memory.

---

17) What is thrashing?

Thrashing is a phenomenon in virtual memory scheme when the processor spends most of its time in swapping pages, rather than executing instructions.

---

18) What are the four necessary and sufficient conditions behind the deadlock?

These are the 4 conditions:

1) Mutual Exclusion Condition: It specifies that the resources involved are non-sharable.

2) Hold and Wait Condition: It specifies that there must be a process that is holding a resource already allocated to it while waiting for additional resource that are currently being held by other processes.

3) No-Preemptive Condition: Resources cannot be taken away while they are being used by processes.

4) Circular Wait Condition: It is an explanation of the second condition. It specifies that the processes in the system form a circular list or a chain where each process in the chain is waiting for a resource held by next process in the chain.

---

19) What is a thread?

A thread is a basic unit of CPU utilization. It consists of a thread ID, program counter, register set and a stack.

---

20) What is FCFS?

FCFS stands for First Come, First Served. It is a type of scheduling algorithm. In this scheme, if a process requests the CPU first, it is allocated to the CPU first. Its implementation is managed by a FIFO queue.

---

21) What is SMP?

SMP stands for Symmetric MultiProcessing. It is the most common type of multiple processor system. In SMP, each processor runs an identical copy of the operating system, and these copies communicate with one another when required.

---

22) What is RAID? What are the different RAID levels?

RAID stands for Redundant Array of Independent Disks. It is used to store the same data redundantly to improve the overall performance.

Following are the different RAID levels:

RAID 0 - Stripped Disk Array without fault tolerance

RAID 1 - Mirroring and duplexing

RAID 2 - Memory-style error-correcting codes

RAID 3 - Bit-interleaved Parity

RAID 4 - Block-interleaved Parity

RAID 5 - Block-interleaved distributed Parity

RAID 6 - P+Q Redundancy

---

23) What is deadlock? Explain.

Deadlock is a specific situation or condition where two processes are waiting for each other to complete so that they can start. But this situation causes hang for both of them.

---

24) Which are the necessary conditions to achieve a deadlock?

There are 4 necessary conditions to achieve a deadlock:

- Mutual Exclusion: At least one resource must be held in a non-sharable mode. If any other process requests this resource, then that process must wait for the resource to be released.

- Hold and Wait: A process must be simultaneously holding at least one resource and waiting for at least one resource that is currently being held by some other process.
- No preemption: Once a process is holding a resource (i.e. once its request has been granted), then that resource cannot be taken away from that process until the process voluntarily releases it.
- Circular Wait: A set of processes { P0, P1, P2, . . ., PN } must exist such that every P[i] is waiting for P[(i + 1) % (N + 1)].

Note: This condition implies the hold-and-wait condition, but it is easier to deal with the conditions if the four are considered separately.

---

25) What is Banker's algorithm?

Banker's algorithm is used to avoid deadlock. It is the one of deadlock-avoidance method. It is named as Banker's algorithm on the banking system where bank never allocates available cash in such a manner that it can no longer satisfy the requirements of all of its customers.

---

26) What is the difference between logical address space and physical address space?

Logical address space specifies the address that is generated by CPU. On the other hand physical address space specifies the address that is seen by the memory unit.

---

27) What is fragmentation?

Fragmentation is a phenomenon of memory wastage. It reduces the capacity and performance because space is used inefficiently.

---

28) How many types of fragmentation occur in Operating System?

There are two types of fragmentation:

- Internal fragmentation: It is occurred when we deal with the systems that have fixed size allocation units.
- External fragmentation: It is occurred when we deal with systems that have variable-size allocation units.

---

29) What is spooling?

Spooling is a process in which data is temporarily gathered to be used and executed by a device, program or the system. It is associated with printing. When different applications send output to the printer at the same time, spooling keeps these all jobs into a disk file and queues them accordingly to the printer.

---

30) What is the difference between internal commands and external commands?

Internal commands are the built-in part of the operating system while external commands are the separate file programs that are stored in a separate folder or directory.

---

31) What is semaphore?

Semaphore is a protected variable or abstract data type that is used to lock the resource being used. The value of the semaphore indicates the status of a common resource.

There are two types of semaphore:

- Binary semaphores
- Counting semaphores

---

32) What is a binary Semaphore?

Binary semaphore takes only 0 and 1 as value and used to implement mutual exclusion and synchronize concurrent processes.

---

33) What is Belady's Anomaly?

Belady's Anomaly is also called FIFO anomaly. Usually, on increasing the number of frames allocated to a process virtual memory, the process execution is faster, because fewer page faults occur. Sometimes, the reverse happens, i.e., the execution time increases even when more frames are allocated to the process. This is Belady's Anomaly. This is true for certain page reference patterns.

---

34) What is starvation in Operating System?

Starvation is Resource management problem. In this problem, a waiting process does not get the resources it needs for a long time because the resources are being allocated to other processes.

---

35) What is aging in Operating System?

Aging is a technique used to avoid the starvation in resource scheduling system.

---

36) What are the advantages of multithreaded programming?

A list of advantages of multithreaded programming:

- Enhance the responsiveness to the users.
- Resource sharing within the process.
- Economical
- Completely utilize the multiprocessing architecture.

---

37) What is the difference between logical and physical address space?

Logical address specifies the address which is generated by the CPU whereas physical address specifies to the address which is seen by the memory unit.

After fragmentation

---

38) What are overlays?

Overlays makes a process to be larger than the amount of memory allocated to it. It ensures that only important instructions and data at any given time are kept in memory.

---

39) When does trashing occur?

Thrashing specifies an instance of high paging activity. This happens when it is spending more time paging instead of executing.

---

40) What is a Batch Operating System?

Batch Operating System is a type of Operating System which creates Batches for the execution of certain jobs or processes.

The Batches contains jobs of such a kind that the jobs or processes which are very similar in the procedure to be followed by the jobs or processes. The Batch Operating System has an operator which performs these tasks.

An operator groups together comparable jobs or processes that have the same criteria into batches. The operator is in charge and takes up the job of grouping jobs or processes with comparable requirements.

Batch Operating System follows First Come First Serve principle.

---

41) Do the Batch Operating System interact with Computer for processing the needs of jobs or processes required?

No, this is not that kind of Operating Systems which tries to interact with the computer. But, this job is taken up by the Operator present in the Batch Operating Systems.

42) What are the advantages of Batch Operating System?

Advantages of Batch Operating System:

1. The time which the Operating System is at rest is very small or also known as Idle Time for the Operating System is very small.
2. Very big tasks can also be managed very easily with the help of Batch Operating Systems
3. Many users can use this Batch Operating Systems.
4. It is incredibly challenging to estimate or determine how long it will take to finish any task. The batch system processors are aware of how long a work will take to complete when it is in line.

---

43) What are the disadvantages of Batch Operating System?

Disadvantages of Batch Operating System:

1. If any work fails in the Batch Operating System, the other jobs will have to wait for an indeterminate period of time.
2. Batch Operating systems are very challenging to debug,
3. Batch Operating Systems can be expensive at times
4. The computer operators who are using Batch Operating Systems must to be knowledgeable with batch systems.

---

44) Where is Batch Operating System used in Real Life

They are used in Payroll System and for generating Bank Statements.

---

45) What is the Function of Operating System?

The most important functions of Operating Systems are:

1. File Management
2. Job Management
3. Process Management
4. Device Management
5. Memory Management

46) What are the Services provided by the Operating System?

The services provided by the Operating Systems are:

1. Security to your computer
2. Protects your computer from external threats
3. File Management
4. Program Execution
5. Helps in Controlling Input Output Devices
6. Useful in Program Creation
7. Helpful in Error Detection
8. Operating System helps in communicating between devices
9. Analyzes the Performance of all devices

47) What is a System Call in Operating Systems?

Programs can communicate with the operating system by making a system call. When a computer application requests anything from the kernel of the operating system, it performs a system call.System call uses Application Programming Interfaces(API)to deliver operating system services to user programs

48) What are the Types of System Calls in Operating Systems?

The System Calls in Operating Systems are:

1. Communication
2. Information Maintenance
3. File Management
4. Device Management
5. Process Contro

49) What are the functions which are present in the Process Control and File Management System Call?

The Functions present in Process Control System Calls are:

1. Create
2. Allocate
3. Abort
4. End
5. Terminate
6. Free Memory

---

50) What are the functions which are present in the File Management System Call?

The Functions present in File Management System Calls are:

1. Create
2. Open
3. Read
4. Close
5. Delete

---

51) What is a Process in Operating Systems?

A process is essentially software that is being run on the Operating System. The Process is a Procedure which must be carried out in a sequential manner.

The fundamental unit of work that has to be implemented in the system is called a process.

An active program known as a process is the basis of all computing. Although relatively similar, the method is not the same as computer code. A process is a "active" entity, in contrast to the program, which is sometimes thought of as some sort of "passive" entity.

---

52) What are the types of Processes in Operating Systems?

The types of Operating System processes are:

1. Operating System Process
2. User Process

---

53) What is Process Control Block (PCB)?

A data structure called a Process Control Block houses details about the processes connected to it. The term "process control block" can also refer to a "task control block," "process table entry," etc.

As data structure for processes is done in terms of the Process Control Block (PCB), it is crucial for process management. Additionally, it describes the operating system's present condition.

---

54) What are the Data Items in Process Control Block?

The Data Items in Process Control Block are:

1. Process State
2. Process Number
3. Program Counter
4. Registers
5. Memory Limits
6. List of Open Files

---

55) What are the Files used in the Process Control Block?

The Files used in Process Control Block are:

1. Central Processing Unit (CPU) Scheduling Information
2. Memory Management Information
3. Accounting Information
4. Input Output Status Information

---

56) What are the differences between Thread and Process

Thread	Process
Threads are executed within the same process	Processes are executed in the different memory spaces

Threads are not independent of each other	Processes are independent of each other

---

57) What are the advantages of Threads in Operating Systems?

The advantages of Threads in Operating System are:

1. Threads are executed very faster than Switches
2. Threads ensure that the communication between threads are very easier
3. The Throughput of the system is increased if the process is divided into multiple threads
4. When a thread in a multi-threaded process completes its execution, its output can be returned right away.

---

58) What are the disadvantages of Threads in Operating Systems?

The disadvantages of Threads in Operating System are:

1. The code becomes more challenging to maintain and debug as there are more threads.
2. The process of creating threads uses up system resources like memory and CPU.
3. Because unhandled exceptions might cause the application to crash, we must manage them inside the worker method.

---

59) What are the types of Threads in Operating System?

The types of Threads in Operating System are:

1. User Level Threads
2. Kernel Level Threads

---

60) What is User Kernel Thread?

The kernel is unaware of the user-level threads since they are implemented at the user level.

They are treated like single-threaded processes under this system. User Kernel Threads are smaller and quicker than kernel-level threads

The User Kernel Threads are represented by a Small Process Control Block (PCB), Stack, Program Counter (PC), Stack.

Here, the User Kernel Threads are independent of Kernel Involvement in Synchronization.

---

61) What are User Kernel Threads Advantages and Disadvantages?

The following are a few advantages of User Kernel Threads:

1. Creating user-level threads is quicker and simpler than creating kernel-level threads. They are also simpler to handle.
2. Any operating system may be used to execute user-level threads.
3. Thread switching in user-level threads does not need kernel mode privileges.

The following are a few drawbacks of User Kernel Threads:

1. Multiprocessing cannot be used effectively by multithreaded applications in user-level threads.
2. If one user-level thread engages in a blocking action, the entire process is halted.

---

62) What is Kernel Level Thread?

Kernel Level Threads are the threads which are handled by the Operating System directly. The kernel controls both the process's threads and the context information for each one. As a result, kernel-level threads execute more slowly than user-level threads.

---

63) What are Kernel Level Threads Advantages and Disadvantages?

The following are some benefits of using kernel-level threads:

1. Kernel-level threads allow the scheduling of many instances of the same process across several CPUs.
2. The kernel functions also support multithreading.
3. Another thread of the same process may be scheduled by the kernel if a kernel-level thread is stalled.

The following list includes several drawbacks of kernel-level threads:

1. To pass control from one thread in a process to another, a mode switch to kernel mode is necessary.
2. Compared to user-level threads, kernel-level threads take longer to create and maintain.

---

64) What is Process Scheduling In Operating Systems?

The task of the process manager that deals with removing the active process from the CPU and choosing a different process based on a certain strategy is known as process scheduling.

An integral component of a multiprogramming operating system is process scheduling. Such operating systems provide the simultaneous loading of numerous processes into the executable memory, each of which utilizes temporal multiplexing to share the CPU.

---

65) What are types of Process Scheduling Techniques in Operating Systems?

The types of Process Scheduling Techniques in Operating Systems are:

1. Pre Emptive Process Scheduling
2. Non Pre Emptive Process Scheduling

---

66) What is Pre Emptive Process Scheduling in Operating Systems?

In this instance of Pre Emptive Process Scheduling, the OS allots the resources to a process for a predetermined period of time. The process transitions from running state to ready state or from waiting state to ready state during resource allocation. This switching happens because the CPU may assign other processes precedence and substitute the currently active process for the higher priority process.

---

67) What is Non Pre Emptive Process Scheduling in Operating Systems?

In this case of Non Pre Emptive Process Scheduling, the resource cannot be withdrawn from a process before the process has finished running. When a running process finishes and transitions to the waiting state, resources are switched.

---

68) What is Context Switching?

Context switching is a technique or approach that the operating system use to move a process from one state to another so that it can carry out its intended function using system CPUs.

When a system performs a switch, it maintains the status of the previous operating process in the form of registers and allots the CPU to the new process to carry out its operations.

The old process must wait in a ready queue while a new one is running in the system. At the point when another process interrupted it, the old process resumes execution.

It outlines the features of an operating system that supports numerous workloads at once without the use of extra processors by allowing several processes to share a single CPU.

---

69) What is the use of Dispatcher in Operating Systems?

After the scheduler completes the process scheduling, a unique application called a dispatcher enters the picture. The dispatcher is the one who moves a process to the desired state or queue once the scheduler has finished its selection task. The module known as the dispatcher is what grants a process control over the CPU once the short-term schedule has chosen it.

---

70) What is the Difference between Dispatcher and Scheduler?

Dispatcher	Scheduler
Dispatcher is the one who moves the process to the desired state	Scheduler is the one which selects a process which is feasible to be executed at this point of time.
The time taken by Dispatcher is known as Dispatch Latency	The Time taken by the Scheduler is not counted basically
Dispatcher is dependent on Scheduler	Scheduler is not dependent of Dispatcher
Dispatcher allows Context Switching to occur	Scheduler only allows the process to the ready queue

---

71) What is Process Synchronization in Operating Systems?

Process synchronization, often known as synchronization, is the method an operating system uses to manage processes that share the same memory space. By limiting the number of processes that may modify the shared memory at once via hardware or variables, it helps ensure the consistency of the data.

---

72) What are the Classical Problems of Process Synchronization?

The Classical Problems of Process Synchronization are:

1. Bound Buffer Problem or Consumer Producer Problem
2. Dining Philosopher's Problem
3. Readers and writers Problem
4. Sleeping Barber Problem

---

73) What is a Peterson's Solution?

Peterson's solution to the critical section issue is a traditional one. The critical section problem makes sure no two processes or jobsalter or modify the value of a resource at the same time.

---

74) What are the Operations in Semaphores?

The Operations in Semaphores are:

1. Wait or P Function ()
2. Signal or V Function ()

---

75) What is a Critical Section Problem?

The section of a program known as the Critical Section attempts to access shared resources.

The operating system has trouble authorizing and preventing processes from entering the crucial part because more than one process cannot operate in the critical area at once.

---

76) What are the methods of Handling Deadlocks?

The methods of handling deadlock are:

1. Deadlock Prevention
2. Deadlock Detection and Recovery
3. Deadlock Avoidance
4. Deadlock Ignorance

---

77) How can we avoid Deadlock?

We can avoid Deadlock by using Banker's Algorithm.

---

78) How can we detect and recover the Deadlock occurred in Operating System?

First, what we need to do is to allow the process to enter the deadlock state. So, it is the time of recovery.

We can recover the process from deadlock state by terminating or aborting all deadlocked processes one at a time.

Process Pre Emption is also another technique used for Deadlocked Process Recovery.

---

79) What is paging in Operating Systems?

Paging is a storage mechanism. Paging is used to retrieve processes from secondary memory to primary memory.

The main memory is divided into small blocks called pages. Now, each of the pages contains the process which is retrieved into main memory and it is stored in one frame of memory.

It is very important to have pages and frames which are of equal sizes which are very useful for mapping and complete utilization of memory.

---

80) What is Address Translation in Paging?

Logical and physical memory addresses, both of which are distinct, are the two types of memory addresses that are employed in the paging process. The logical address is the address that the CPU creates for each page in the secondary memory, but the physical address is the actual location of the frame where each page will be allocated. We now require a technique known as address translation carried out by the page table in order to translate this logical address into a physical address.

81) What is Translational Look Aside Buffer?

Whenever logical address is created by the Central Processing Unit (CPU), the page number is stored in the Translational Look Aside Buffer. Along, with the page number, the frame number is also stored.

82) What are Page Replacement Algorithms in Operating Systems?

The Page Replacement Algorithms in Operating Systems are:

1. First In First Out
2. Optimal
3. Least Recently Used
4. Most Recently Used

83) In which Page Replacement Algorithm does Belady's Anomaly occur?

In First in First out Page Replacement Algorithm Belady's Anomaly occurs.

84) What are Process Scheduling Algorithms in Operating System?

The Process Scheduling Algorithms in Operating Systems are:

1. First Come First Serve CPU Scheduling Algorithm
2. Priority Scheduling CPU Scheduling Algorithm
3. Shortest Job First CPU Scheduling Algorithm
4. Round Robin CPU Scheduling Algorithm
5. Longest Job First CPU Scheduling Algorithm
6. Shortest Remaining Time First CPU Scheduling Algorithm
7. Multiple Queue CPU Scheduling Algorithm

85) What is Round Robin CPU Scheduling Algorithm?

Round Robin is a CPU scheduling mechanism whose cycles around assigning each task a specific time slot. It is the First come First Serve CPU Scheduling method prior Pre Emptive Process Scheduling approach. The Round Robin CPU algorithm frequently emphasizes the Time Sharing method.

---

86) What is Disk Scheduling in Operating Systems?

Operating systems use disk scheduling to plan when Input or Output requests for the disk will arrive. Input or Output scheduling is another name for disk scheduling.

---

87) What is the importance of Disk Scheduling in Operating Systems?

1. One Input or Output request may be fulfilled by the disk controller at a time, even if several Input or Output requests could come in from other processes. Other Input or Output requests must thus be scheduled and made to wait in the waiting queue.
2. The movement of the disk arm might increase if two or more requests are placed far apart from one another.
3. Since hard disks are among the slower components of the computer system, they must be accessible quickly.

---

88) What are the Disk Scheduling Algorithms used in Operating Systems?

The Disk Scheduling Algorithms used in Operating Systems are:

1. First Come First Serve
2. Shortest Seek Time First
3. LOOK
4. SCAN
5. C SCAN
6. C LOOK

---

89) What is Monitors in the context of Operating Systems?

A feature of programming languages called monitors and ithelps in controlledaccess to shared data. The Monitor is a collection of shared actions, data structures, and synchronization between parallel procedure calls. A monitor is therefore sometimes referred to as a synchronization tool. Some of the languages that support the usage of monitors are Java, C#, Visual Basic, Ada, and concurrent Euclid. Although other processes cannot access the monitor's internal variables, they can invoke its methods

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Networking Interview Questions

A list of top frequently asked Networking interview questions and answers are given below.

---

1) What is the network?

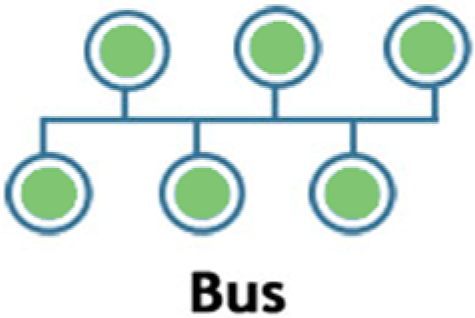
- A network is a set of devices that are connected with a physical media link. In a network, two or more nodes are connected by a physical link or two or more networks are connected by one or more nodes.
- A network is a collection of devices connected to each other to allow the sharing of data.
- Example of a network is an internet. An internet connects the millions of people across the world.

---

2) What do you mean by network topology?

Network topology specifies the layout of a computer network. It shows how devices and cables are connected to each other. The types of topologies are:

Bus:



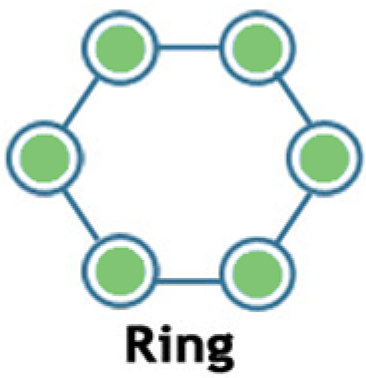
- Bus topology is a network topology in which all the nodes are connected to a single cable known as a central cable or bus.
- It acts as a shared communication medium, i.e., if any device wants to send the data to other devices, then it will send the data over the bus which in turn sends the data to all the attached devices.
- Bus topology is useful for a small number of devices. As if the bus is damaged then the whole network fails.

Star:

- Star topology is a network topology in which all the nodes are connected to a single device known as a central device.
- Star topology requires more cable compared to other topologies. Therefore, it is more robust as a failure in one cable will only disconnect a specific computer connected to this cable.
- If the central device is damaged, then the whole network fails.
- Star topology is very easy to install, manage and troubleshoot.

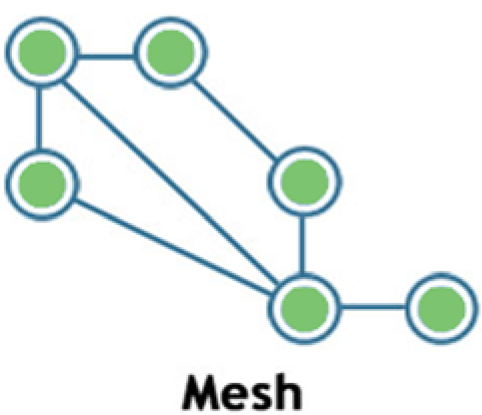
- Star topology is commonly used in office and home networks.

Ring



- Ring topology is a network topology in which nodes are exactly connected to two or more nodes and thus, forming a single continuous path for the transmission.
- It does not need any central server to control the connectivity among the nodes.
- If the single node is damaged, then the whole network fails.
- Ring topology is very rarely used as it is expensive, difficult to install and manage.
- Examples of Ring topology are SONET network, SDH network, etc.

Mesh



- Mesh topology is a network topology in which all the nodes are individually connected to other nodes.
- It does not need any central switch or hub to control the connectivity among the nodes.
- Mesh topology is categorized into two parts:
    - Fully connected mesh topology: In this topology, all the nodes are connected to each other.
    - Partially connected mesh topology: In this topology, all the nodes are not connected to each other.
- It is a robust as a failure in one cable will only disconnect the specified computer connected to this cable.
- Mesh topology is rarely used as installation and configuration are difficult when connectivity gets more.
- Cabling cost is high as it requires bulk wiring.

Tree

- Tree topology is a combination of star and bus topology. It is also known as the expanded star topology.
- In tree topology, all the star networks are connected to a single bus.
- Ethernet protocol is used in this topology.
- In this, the whole network is divided into segments known as star networks which can be easily maintained. If one segment is damaged, but there is no effect on other segments.
- Tree topology depends on the "main bus," and if it breaks, then the whole network gets damaged.

Hybrid

- A hybrid topology is a combination of different topologies to form a resulting topology.
- If star topology is connected with another star topology, then it remains star topology. If star topology is connected with different topology, then it becomes a Hybrid topology.
- It provides flexibility as it can be implemented in a different network environment.
- The weakness of a topology is ignored, and only strength will be taken into consideration.

---

3) What are the advantages of Distributed Processing?

A list of advantages of distributed processing:

- Secure
- Support Encapsulation
- Distributed database
- Faster Problem solving
- Security through redundancy
- Collaborative Processing

---

4) What is the criteria to check the network reliability?

Network reliability: Network reliability means the ability of the network to carry out the desired operation through a network such as communication through a network.

Network reliability plays a significant role in the network functionality. The network monitoring systems and devices are the essential requirements for making the network reliable.The network monitoring system identifies the problems that are occurred in the network while the network devices ensure that data should reach the appropriate destination.

The reliability of a network can be measured by the following factors:

- Downtime: The downtime is defined as the required time to recover.
- Failure Frequency: It is the frequency when it fails to work the way it is intended.
- Catastrophe: It indicates that the network has been attacked by some unexpected event such as fire, earthquake.

---

5) Which are the different factors that affect the security of a network?

There are mainly two security affecting factors:

- Unauthorized Access
- Viruses
-

---

6) Which are the different factors that affect the reliability of a network?

The following factors affect the reliability of a network:

- Frequency of failure
- Recovery time of a network after a failure

---

7) Which are the different factors that affect the performance of a network?

The following factors affect the performance of a network:

- Large number of users
- Transmission medium types
- Hardware
- Software

8) What makes a network effective and efficient?

There are mainly two criteria which make a network effective and efficient:

- Performance: : performance can be measured in many ways like transmit time and response time.
- Reliability: reliability is measured by frequency of failure.
- Robustness: robustness specifies the quality or condition of being strong and in good condition.
- Security: It specifies how to protect data from unauthorized access and viruses.

---

9) What is bandwidth?

Every signal has a limit of upper range frequency and lower range frequency. The range of limit of network between its upper and lower frequency is called bandwidth.

---

10) What is a node and link?

A network is a connection setup of two or more computers directly connected by some physical mediums like optical fiber or coaxial cable. This physical medium of connection is known as a link, and the computers that it is connected are known as nodes.

---

11) What is a gateway? Is there any difference between a gateway and router?

A node that is connected to two or more networks is commonly known as a gateway. It is also known as a router. It is used to forward messages from one network to another. Both the gateway and router regulate the traffic in the network.

Differences between gateway and router:

A router sends the data between two similar networks while gateway sends the data between two dissimilar networks.

---

12) What is DNS?

DNS is an acronym stands for Domain Name System.

- DNS was introduced by Paul Mockapetris and Jon Postel in 1983.
- It is a naming system for all the resources over the internet which includes physical nodes and applications. It is used to locate to resource easily over a network.
- DNS is an internet which maps the domain names to their associated IP addresses.
- Without DNS, users must know the IP address of the web page that you wanted to access.

Working of DNS:

If you want to visit the website of "javaTpoint", then the user will type "https://www.javatpoint.com" into the address bar of the web browser. Once the domain name is entered, then the domain name system will translate the domain name into the IP address which can be easily interpreted by the computer. Using the IP address, the computer can locate the web page requested by the user.

---

13) What is DNS forwarder?

- A forwarder is used with DNS server when it receives DNS queries that cannot be resolved quickly. So it forwards those requests to external DNS servers for resolution.
- A DNS server which is configured as a forwarder will behave differently than the DNS server which is not configured as a forwarder.
- Following are the ways that the DNS server behaves when it is configured as a forwarder:

- When the DNS server receives the query, then it resolves the query by using a cache.
- If the DNS server is not able to resolve the query, then it forwards the query to another DNS server.
- If the forwarder is not available, then it will try to resolve the query by using root hint.

---

14) What is NIC?

- NIC stands for Network Interface Card. It is a peripheral card attached to the PC to connect to a network. Every NIC has its own MAC address that identifies the PC on the network.
- It provides a wireless connection to a local area network.
- NICs were mainly used in desktop computers.

---

15) What is the meaning of 10Base-T?

It is used to specify data transfer rate. In 10Base-T, 10 specify the data transfer rate, i.e., 10Mbps. The word Base specifies the baseband as opposed to broadband. T specifies the type of the cable which is a twisted pair.

---

16) What is NOS in computer networking?

- NOS stands for Network Operating System. It is specialized software which is used to provide network connectivity to a computer to make communication possible with other computers and connected devices.
- NOS is the software which allows the device to communicate, share files with other devices.
- The first network operating system was Novel NetWare released in 1983. Some other examples of NOS are Windows 2000, Windows XP, Linux, etc.

---

17) What are the different types of networks?

Networks can be divided on the basis of area of distribution. For example:

- PAN (Personal Area Network): Its range limit is up to 10 meters. It is created for personal use. Generally, personal devices are connected to this network. For example computers, telephones, fax, printers, etc.
- LAN (Local Area Network): It is used for a small geographical location like office, hospital, school, etc.
- HAN (House Area Network): It is actually a LAN that is used within a house and used to connect homely devices like personal computers, phones, printers, etc.
- CAN (Campus Area Network): It is a connection of devices within a campus area which links to other departments of the organization within the same campus.
- MAN (Metropolitan Area Network): It is used to connect the devices which span to large cities like metropolitan cities over a wide geographical area.
- WAN (Wide Area Network): It is used over a wide geographical location that may range to connect cities and countries.
- GAN (Global Area Network): It uses satellites to connect devices over global are.

---

18) What is POP3?

POP3 stands for Post Office Protocol version3. POP is responsible for accessing the mail service on a client machine. POP3 works on two models such as Delete mode and Keep mode.

---

19) What do you understand by MAC address?

MAC stands for Media Access Control. It is the address of the device at the Media Access Control Layer of Network Architecture. It is a unique address means no two devices can have same MAC addresses.

---

20) What is IP address?

IP address is a unique 32 bit software address of a computer in a network system.

---

21) What is private IP address?

There are three ranges of IP addresses that have been reserved for IP addresses. They are not valid for use on the internet. If you want to access internet on these private IPs, you must have to use proxy server or NAT server.

---

22) What is public IP address?

A public IP address is an address taken by the Internet Service Provider which facilitates you to communication on the internet.

---

23) What is APIPA?

APIPA is an acronym stands for Automatic Private IP Addressing. This feature is generally found in Microsoft operating system.

---

24) What is the full form of ADS?

- ADS stands for Active Directory Structure.
- ADS is a microsoft technology used to manage the computers and other devices.
- ADS allows the network administrators to manage the domains, users and objects within the network.
- ADS consists of three main tiers:
    - Domain: Users that use the same database will be grouped into a single domain.
    - Tree: Multiple domains can be grouped into a single tree.
    - Forest: Multiple trees can be grouped into a single forest.

---

25) What is RAID?

RAID is a method to provide Fault Tolerance by using multiple Hard Disc Drives.

---

26) What is anonymous FTP?

Anonymous FTP is used to grant users access to files in public servers. Users which are allowed access to data in these servers do not need to identify themselves, but instead log in as an anonymous guest.

---

27) What is protocol?

A protocol is a set of rules which is used to govern all the aspects of information communication.

---

28) What are the main elements of a protocol?

The main elements of a protocol are:

- Syntax: It specifies the structure or format of the data. It also specifies the order in which they are presented.
- Semantics: It specifies the meaning of each section of bits.
- Timing: Timing specifies two characteristics: When data should be sent and how fast it can be sent.

---

29 What is the Domain Name System?

There are two types of client/server programs. First is directly used by the users and the second supports application programs.

The Domain Name System is the second type supporting program that is used by other programs such as to find the IP address of an e-mail recipient.

---

30) What is link?

A link is connectivity between two devices which includes the cables and protocols used in order to make communication between devices.

---

31) How many layers are in OSI reference model?

OSI reference model: OSI reference model is an ISO standard which defines a networking framework for implementing the protocols in seven layers. These seven layers can be grouped into three categories:

- Network layer: Layer 1, Layer 2 and layer 3 are the network layers.
- Transport layer: Layer 4 is a transport layer.
- Application layer. Layer 5, Layer 6 and Layer 7 are the application layers.

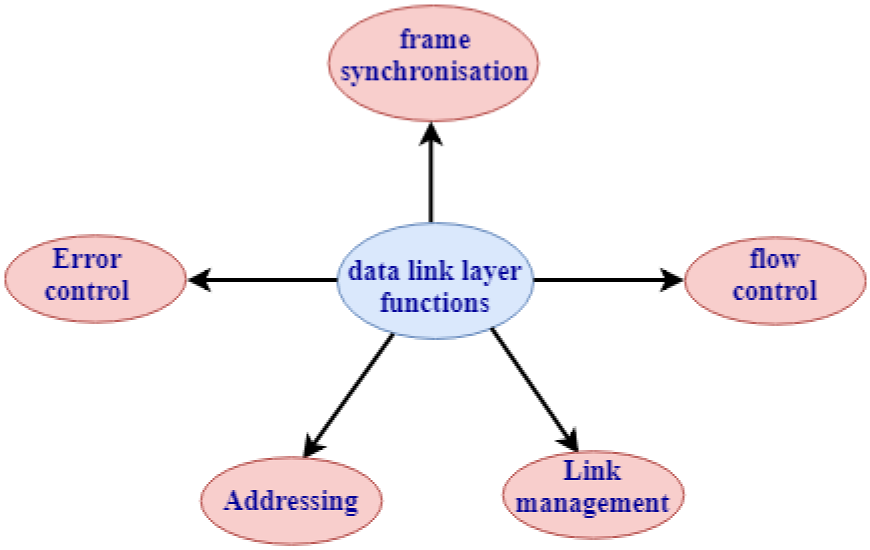
There are 7 layers in the OSI reference model.

1. Physical Layer

- It is the lowest layer of the OSI reference model.
- It is used for the transmission of an unstructured raw bit stream over a physical medium.
- Physical layer transmits the data either in the form of electrical/optical or mechanical form.
- The physical layer is mainly used for the physical connection between the devices, and such physical connection can be made by using twisted-pair cable, fibre-optic or wireless transmission media.

2. DataLink Layer

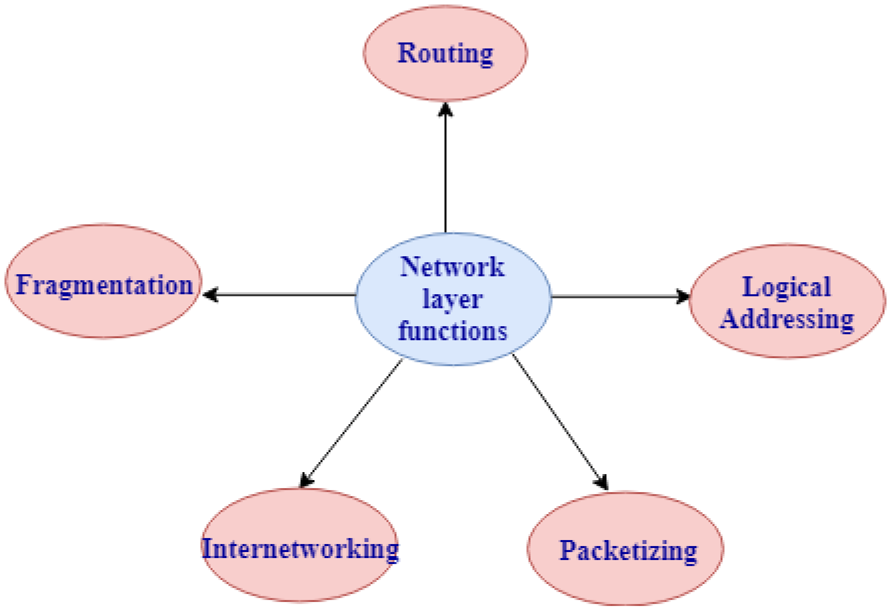
- It is used for transferring the data from one node to another node.
- It receives the data from the network layer and converts the data into data frames and then attach the physical address to these frames which are sent to the physical layer.
- It enables the error-free transfer of data from one node to another node.
  Functions of Data-link layer:



- Frame synchronization: Data-link layer converts the data into frames, and it ensures that the destination must recognize the starting and ending of each frame.
- Flow control: Data-link layer controls the data flow within the network.
- Error control: It detects and corrects the error occurred during the transmission from source to destination.
- Addressing: Data-link layer attach the physical address with the data frames so that the individual machines can be easily identified.
- Link management: Data-link layer manages the initiation, maintenance and, termination of the link between the source and destination for the effective exchange of data.

3. Network Layer

- Network layer converts the logical address into the physical address.
- It provides the routing concept means it determines the best route for the packet to travel from source to the destination.
  Functions of network layer:



- Routing: The network layer determines the best route from source to destination. This function is known as routing.
- Logical addressing: The network layer defines the addressing scheme to identify each device uniquely.
- Packetizing: The network layer receives the data from the upper layer and converts the data into packets. This process is known as packetizing.
- Internetworking: The network layer provides the logical connection between the different types of networks for forming a bigger network.
- Fragmentation: It is a process of dividing the packets into the fragments.

4. Transport Layer

- It delivers the message through the network and provides error checking so that no error occurs during the transfer of data.
- It provides two kinds of services:
    - Connection-oriented transmission: In this transmission, the receiver sends the acknowledgement to the sender after the packet has been received.
    - Connectionless transmission: In this transmission, the receiver does not send the acknowledgement to the sender.

5. Session Layer

- The main responsibility of the session layer is beginning, maintaining and ending the communication between the devices.
- Session layer also reports the error coming from the upper layers.
- Session layer establishes and maintains the session between the two users.

6. Presentation Layer

- The presentation layer is also known as a Translation layer as it translates the data from one format to another format.
- At the sender side, this layer translates the data format used by the application layer to the common format and at the receiver side, this layer translates the common format into a format used by the application layer.
  Functions of presentation layer:
  - Character code translation
  - Data conversion
  - Data compression
  - Data encryption

7. Application Layer

- Application layer enables the user to access the network.
- It is the topmost layer of the OSI reference model.
- Application layer protocols are file transfer protocol, simple mail transfer protocol, domain name system, etc.
- The most widely used application protocol is HTTP(Hypertext transfer protocol). A user sends the request for the web page using HTTP.

---

32) What is the usage of OSI physical layer?

The OSI physical layer is used to convert data bits into electrical signals and vice versa. On this layer, network devices and cable types are considered and setup.

---

33) Explain the functionality of OSI session layer?

OSI session layer provides the protocols and means for two devices on the network to communicate with each other by holding a session. This layer is responsible for setting up the session, managing information exchange during the session, and tear-down process upon termination of the session.

---

34) What is the maximum length allowed for a UTP cable?

The maximum length of UTP cable is 90 to 100 meters.

---

35) What is RIP?

- RIP stands for Routing Information Protocol. It is accessed by the routers to send data from one network to another.
- RIP is a dynamic protocol which is used to find the best route from source to the destination over a network by using the hop count algorithm.
- Routers use this protocol to exchange the network topology information.
- This protocol can be used by small or medium-sized networks.

---

36) What do you understand by TCP/IP?

TCP/IP is short for Transmission Control Protocol /Internet protocol. It is a set of protocol layers that is designed for exchanging data on different types of networks.

---

37) What is netstat?

The "netstat" is a command line utility program. It gives useful information about the current TCP/IP setting of a connection.

---

38) What do you understand by ping command?

The "ping" is a utility program that allows you to check the connectivity between the network devices. You can ping devices using its IP address or name.

---

39) What is Sneakernet?

Sneakernet is the earliest form of networking where the data is physically transported using removable media.

---

40) Explain the peer-peer process.

The processes on each machine that communicate at a given layer are called peer-peer process.

---

41) What is a congested switch?

A switch receives packets faster than the shared link. It can accommodate and stores in its memory, for an extended period of time, then the switch will eventually run out of buffer space, and some packets will have to be dropped. This state is called a congested state.

---

42) What is multiplexing in networking?

In Networking, multiplexing is the set of techniques that is used to allow the simultaneous transmission of multiple signals across a single data link.

---

43) What are the advantages of address sharing?

Address sharing provides security benefit instead of routing. That's because host PCs on the Internet can only see the public IP address of the external interface on the computer that provides address translation and not the private IP addresses on the internal network.

---

44) What is RSA Algorithm?

RSA is short for Rivest-Shamir-Adleman algorithm. It is mostly used for public key encryption.

---

45) How many layers are in TCP/IP?

There are basic 4 layers in TCP/IP:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Layer

---

46) What is the difference between TCP/IP model and the OSI model?

Following are the differences between the TCP/IP model and OSI model:

Full form of TCP is transmission control protocol.	Full form of OSI is Open System Interconnection.
TCP/IP has 4 layers.	OSI has 7 layers.
TCP/IP is more reliable than the OSI model.	OSI model is less reliable as compared to the TCP/IP model.
TCP/IP model uses horizontal approach.	OSI model uses vertical approach.

TCP/IP model uses both session and presentation layer in the application layer.	OSI Reference model uses separate session and presentation layers.
TCP/IP model developed the protocols first and then model.	OSI model developed the model first and then protocols.
In Network layer, TCP/IP model supports only connectionless communication.	In the Network layer, the OSI model supports both connection-oriented and connectionless communication.
TCP/IP model is a protocol dependent.	OSI model is a protocol independent.

---

47) What is the difference between domain and workgroup?

A workgroup is a peer-to-peer computer network.	A domain is a Client/Server network.
A Workgroup can consist of maximum 10 computers.	A domain can consist up to 2000 computers.
Every user can manage the resources individually on their PCs.	There is one administrator to administer the domain and its resources.

All the computers must be on the same local area network.	The computer can be on any network or anywhere in the world.
Each computer must be changed manually.	Any change made to the computer will reflect the changes to all the computers.

OOPs Interview Questions

A list of top frequently asked OOPs interview questions and answers are given below.

1) What do you understand by OOP?

OOP stands for object-oriented programming. It is a programming paradigm that revolves around the object rather than function and procedure. In other words, it is an approach for developing applications that emphasize on objects. An object is a real word entity that contains data and code. It allows binding data and code together.

2) Name any seven widely used OOP languages.

There are various OOP languages but the most widely used are:

- Python
- Java
- Go
- Dart
- C++
- C#
- Ruby

3) What is the purpose of using OOPs concepts?

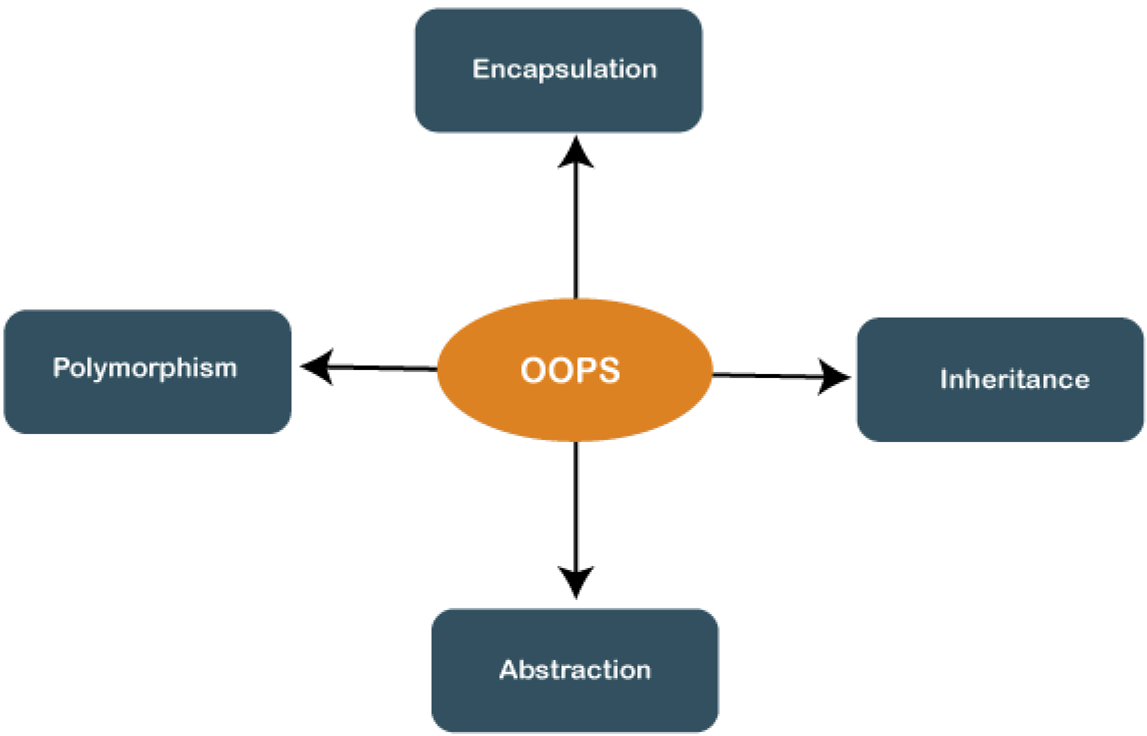
The aim of OOP is to implement real-world entities like inheritance, hiding, polymorphism in programming. The main purpose of OOP is to bind together the data and the functions that operate on them so that no other part of the code can access this data except that function.

---

4) What are the four main features of OOPs?

The OOP has the following four features:

- Inheritance
- Encapsulation
- Polymorphism
- Data Abstraction



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5) Why OOP is so popular?

OOPs, programming paradigm is considered as a better style of programming. Not only it helps in writing a complex piece of code easily, but it also allows users to handle and maintain them easily as well. Not only that, the main pillar of OOPs

- Data Abstraction, Encapsulation, Inheritance, and Polymorphism, makes it easy for programmers to solve complex scenarios. As a result of these, OOPs is so popular.

---

6) What are the advantages and disadvantages of OOP?

Advantages of OOP

- It follows a bottom-up approach.
- It models the real word well.
- It allows us the reusability of code.
- Avoids unnecessary data exposure to the user by using the abstraction.
- OOP forces the designers to have a long and extensive design phase that results in better design and fewer flaws.
- Decompose a complex problem into smaller chunks.
- Programmer are able to reach their goals faster.
- Minimizes the complexity.
- Easy redesign and extension of code that does not affect the other functionality.

Disadvantages of OOP

- Proper planning is required.
- Program design is tricky.
- Programmer should be well skilled.
- Classes tend to be overly generalized.

---

7) What are the limitations of OOPs?

- Requires intensive testing processes.
- Solving problems takes more time as compared to Procedure Oriented Programming.
- The size of the programs created using this approach may become larger than the programs written using the procedure-oriented programming approach.

- Software developed using this approach requires a substantial amount of pre-work and planning.
- OOP code is difficult to understand if you do not have the corresponding class documentation.
- In certain scenarios, these programs can consume a large amount of memory.
- Not suitable for small problems.
- Takes more time to solve problems.

---

8) What are the differences between object-oriented programming and structural programming?

It follows a bottom-up approach.	It follows a top-down approach.
It provides data hiding.	Data hiding is not allowed.
It is used to solve complex problems.	It is used to solve moderate problems.
It allows reusability of code that reduces redundancy of code.	Reusability of code is not allowed.
It is based on objects rather than functions and procedures.	It provides a logical structure to a program in which the program is divided into functions.
It provides more security as it has a data hiding feature.	It provides less security as it does not support the data hiding feature.

More abstraction more flexibility.	Less abstraction less flexibility.
It focuses on data.	It focuses on the process or logical structure.

---

9) What do you understand by pure object-oriented language? Why Java is not a pure object-oriented programming language?

The programming language is called pure object-oriented language that treats everything inside the program as an object. The primitive types are not supported by the pure OOPs language. There are some other features that must satisfy by a pure object-oriented language:

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- All predefined types are objects
- All user-defined types are objects
- All operations performed on objects must be only through methods exposed to the objects.

Java is not a pure object-oriented programming language because pre-defined data types in Java are not treated as objects. Hence, it is not an object-oriented language.

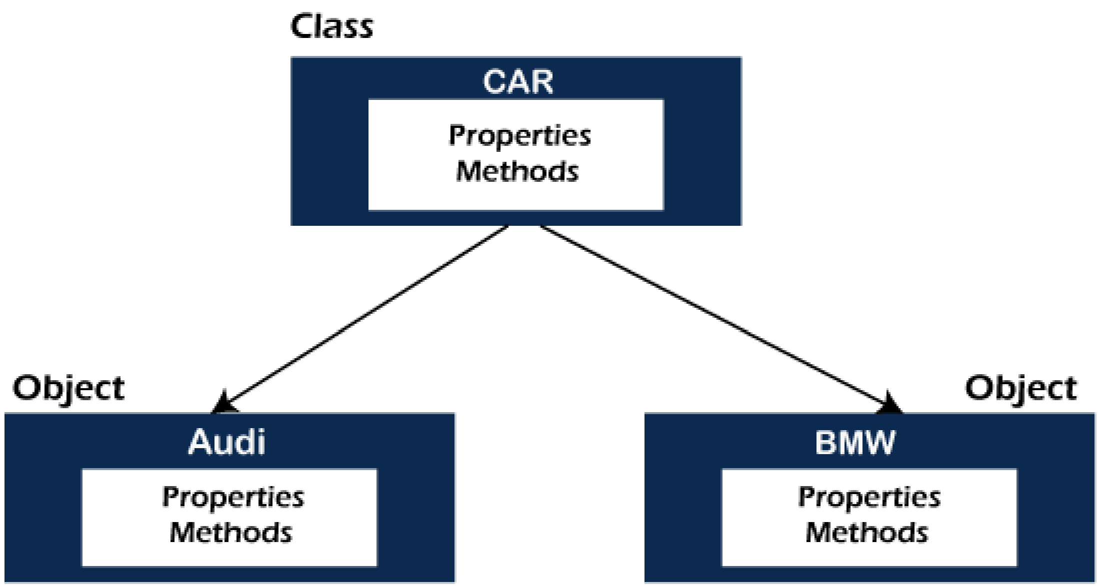
---

10) What do you understand by class and object? Also, give example.

Class: A class is a blueprint or template of an object. It is a user-defined data type. Inside a class, we define variables, constants, member functions, and other functionality. It does not consume memory at run time. Note that classes are not considered as a data structure. It is a logical entity. It is the best example of data binding.

Object: An object is a real-world entity that has attributes, behavior, and properties. It is referred to as an instance of the class. It contains member functions, variables that we have defined in the class. It occupies space in the memory. Different objects have different states or attributes, and behaviors.

The following figure best illustrates the class and object.



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11) What are the differences between class and object?

It is a logical entity.	It is a real-world entity.
It is conceptual.	It is real.
It binds data and methods together into a single unit.	It is just like a variable of a class.

It does not occupy space in the memory.	It occupies space in the memory.
It is a data type that represents the blueprint of an object.	It is an instance of the class.
It is declared once.	Multiple objects can be declared as and when required.
It uses the keyword class when declared.	It uses the new keyword to create an object.
A class can exist without any object.	Objects cannot exist without a class.

---

12) What are the key differences between class and structure?

Class is a group of common objects that shares common properties.	The structure is a collection of different data types.
It deals with data members and member functions.	It deals with data members only.
It supports inheritance.	It does not support inheritance.
Member variables cannot be initialized directly.	Member variables can be initialized directly.

It is of type reference.	It is of a type value.
It's members are private by default.	It's members are public by default.
The keyword class defines a class.	The keyword struct defines a structure.
An instance of a class is an object.	An instance of a structure is a structure variable.
Useful while dealing with the complex data structure.	Useful while dealing with the small data structure.

---

13) What is the concept of access specifiers when should we use these?

In OOPs language, access specifiers are reserved keyword that is used to set the accessibility of the classes, methods and other members of the class. It is also known as access modifiers. It includes public, private, and protected. There is some other access specifier that is language-specific. Such as Java has another access specifier default. These access specifiers play a vital role in achieving one of the major functions of OOP, i.e. encapsulation. The following table depicts the accessibility.

Specifiers	Within Same Class	In Derived Class	Outside the Class
Private	Yes	No	No
Protected	Yes	Yes	No
Public	Yes	Yes	Yes

---

14) What are the manipulators in OOP and how it works?

Manipulators are helping functions. It is used to manipulate or modify the input or output stream. The modification is possible by using the insertion (<<) and extraction (>>) operators. Note that the modification of input or output stream does not mean to change the values of variables. There are two types of manipulators with arguments or without arguments.

The example of manipulators that do not have arguments is endl, ws, flush, etc. Manipulators with arguments are setw(val), setfill(c), setbase(val), setiosflags(flag). Some other manipulators are showpos, fixed, scientific, hex, dec, oct, etc.

---

15) What are the rules for creating a constructor?

- It cannot have a return type.
- It must have the same name as the Class name.
- It cannot be marked as static.
- It cannot be marked as abstract.
- It cannot be overridden.
- It cannot be final.

16) What are the differences between the constructor and the method in Java?

Constructor has the same name as the class name.	The method name and class name are not the same.
It is a special type of method that is used to initialize an object of its class.	It is a set of instructions that can be invoked at any point in a program.
It creates an instance of a class.	It is used to execute Java code.
It is invoked implicitly when we create an object of the class.	It gets executed when we explicitly called it.
It cannot be inherited by the subclass.	It can be inherited by the subclass.
It does not have any return type.	It must have a return type.
It cannot be overridden in Java.	It can be overridden in Java.
It cannot be declared as static.	It can be declared as static.
Java compiler automatically provides a default constructor.	Java compiler does not provide any method by default.

17) How does procedural programming be different from OOP differ?

It is based on functions.	It is based on real-world objects.
It follows a top-down approach.	It follows a bottom-up approach.
It is less secure because there is no proper way to hide data.	It provides more security.
Data is visible to the whole program.	It encapsulates the data.
Reuse of code is not allowed.	The code can be reused.
Modification and extension of code are not easy.	We can easily modify and extend code.
Examples of POP are C, VB, FORTRAN, Pascal, etc.	Examples of OOPs are C++, Java, C#, .NET, etc.

---

18) What are the differences between error and exception?

Recoverable/ Irrecoverable	Exception can be recovered by using the try-catch block.	An error cannot be recovered.
Type	It can be classified into two categories i.e. checked and unchecked.	All errors in Java are unchecked.
Occurrence	It occurs at compile time or run time.	It occurs at run time.
Package	It belongs to java.lang.Exception package.	It belongs to java.lang.Error package.
Known or unknown	Only checked exceptions are known to the compiler.	Errors will not be known to the compiler.
Causes	It is mainly caused by the application itself.	It is mostly caused by the environment in which the application is running.
Example	Checked Exceptions: SQLException, IOException Unchecked Exceptions: ArrayIndexOutOfBoundException, NullPointerException, ArithmaticException	Java.lang.StackOverFlow, java.lang.OutOfMemoryError

19) What are the characteristics of an abstract class?

An abstract class is a class that is declared as abstract. It cannot be instantiated and is always used as a base class. The characteristics of an abstract class are as follows:

- Instantiation of an abstract class is not allowed. It must be inherited.
- An abstract class can have both abstract and non-abstract methods.
- An abstract class must have at least one abstract method.
- You must declare at least one abstract method in the abstract class.
- It is always public.
- It is declared using the abstract

The purpose of an abstract class is to provide a common definition of the base class that multiple derived classes can share.

---

20) Is it possible for a class to inherit the constructor of its base class?

No, a class cannot inherit the constructor of its base class.

---

21) Identify which OOPs concept should be used in the following scenario?

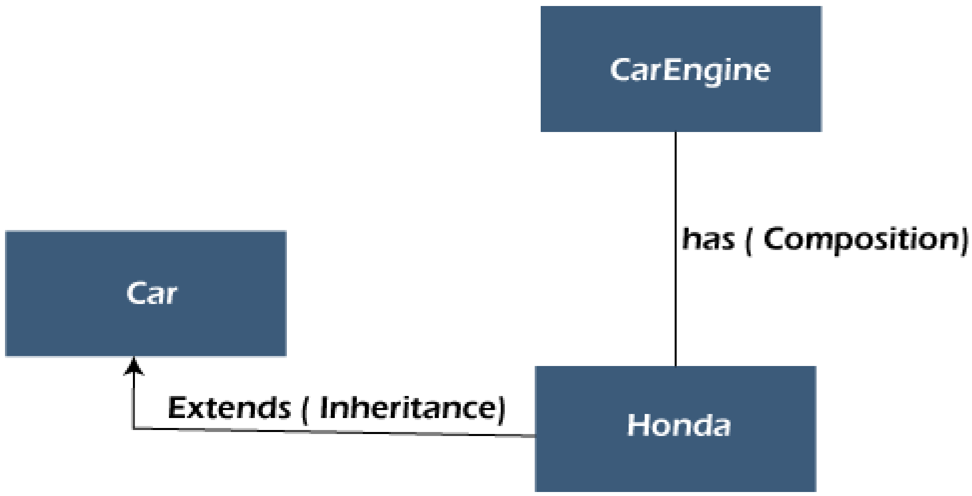
A group of 5 friends, one boy never gives any contribution when the group goes for the outing. Suddenly a beautiful girl joins the same group. The boy who never contributes is now spending a lot of money for the group.

Runtime Polymorphism

---

22) What is composition?

Composition is one of the vital concepts in OOP. It describes a class that references one or more objects of other classes in instance variables. It allows us to model a has-a association between objects. We can find such relationships in the real world. For example, a car has an engine. the following figure depicts the same



The main benefits of composition are:

- Reuse existing code
- Design clean APIs
- Change the implementation of a class used in a composition without adapting any external clients.

---

23) What are the differences between copy constructor and assignment operator?

The copy constructor and the assignment operator (=) both are used to initialize one object using another object. The main difference between the two is that the copy constructor allocates separate memory to both objects i.e. existing object and newly created object while the assignment operator does not allocate new memory for the newly created object. It uses the reference variable that points to the previous memory block (where an old object is located).

Syntax of Copy Constructor

```
class_name (const class_name &obj)
{
//body
}
```

Syntax of Assignment Operator

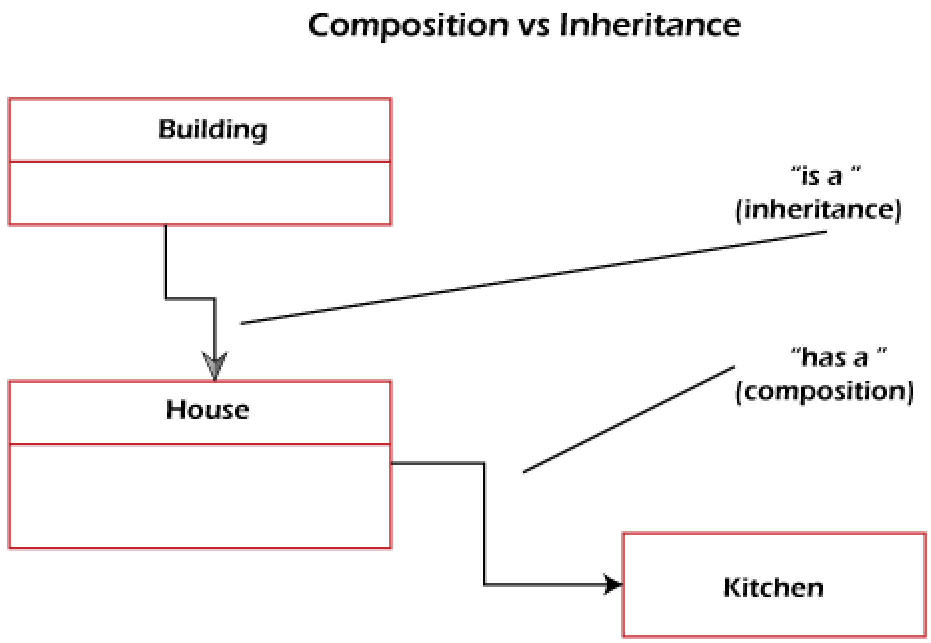
```
class_name obj1, obj2;
obj1=obj2;
```

It is an overloaded constructor.	It is an operator.
It creates a new object as a copy of an existing object.	It assigns the value of one object to another object both of which already exist.
The copy constructor is used when a new object is created with some existing object.	It is used when we want to assign an existing object to a new object.
Both the objects use separate memory locations.	Both objects share the same memory but use the two different reference variables that point to the same location.
If no copy constructor is defined in the class, the compiler provides one.	If the assignment operator is not overloaded then the bitwise copy will be made.

---

24) What is the difference between Composition and Inheritance?

Inheritance means an object inheriting reusable properties of the base class. Compositions mean that an object holds other objects. In Inheritance, there is only one object in memory (derived object) whereas, in Composition, the parent object holds references of all composed objects. From a design perspective, inheritance is "is a" relationship among objects whereas Composition is "has a" relationship among objects.



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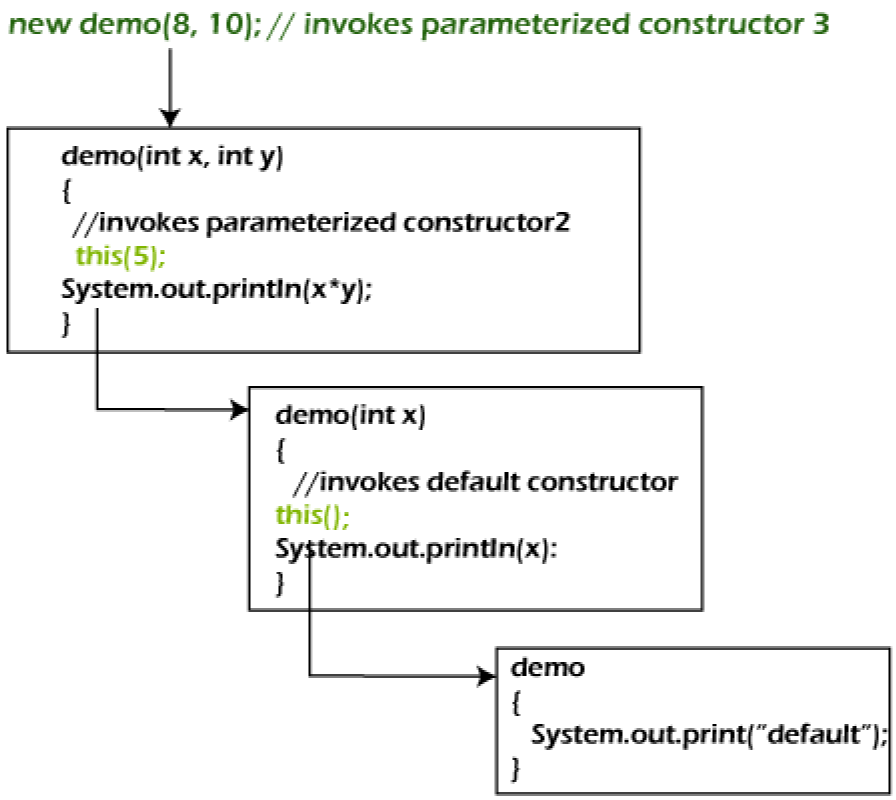
25) What is constructor chaining?

In OOPs, constructor chaining is a sequence of invoking constructors (of the same class) upon initializing an object. It is used when we want to invoke a number of constructors, one after another by using only an instance. In other words, if a class has more than one constructor (overloaded) and one of them tries to invoke another constructor, this process is known as constructor chaining. In C++, it is known as constructor delegation and it is present from C++ 11.



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26) What are the limitations of inheritance?

- The main disadvantage of using inheritance is two classes get tightly coupled. That means one cannot be used independently of the other. If a method or aggregate is deleted in the Super Class, we have to refactor using that method in SubClass.
- Inherited functions work slower compared to normal functions.
- Need careful implementation otherwise leads to improper solutions.

---

27) What are the differences between Inheritance and Polymorphism?

Inheritance is one in which a derived class inherits the already existing class's features.	Polymorphism is one that you can define in different forms.
It refers to using the structure and behavior of a superclass in a subclass.	It refers to changing the behavior of a superclass in the subclass.
It is required in order to achieve polymorphism.	In order to achieve polymorphism, inherence is not required.
It is applied to classes.	It is applied to functions and methods.
It can be single, hybrid, multiple, hierarchical, multipath, and multilevel inheritance.	There are two types of polymorphism compile time and run time.
It supports code reusability and reduces lines of code.	It allows the object to decide which form of the function to be invoked at run-time (overriding) and compile-time (overloading).

---

28) What is Coupling in OOP and why it is helpful?

In programming, separation of concerns is known as coupling. It means that an object cannot directly change or modify the state or behavior of other objects. It defines how closely two objects are connected together. There are two types of coupling, loose coupling, and tight coupling.

Objects that are independent of one another and do not directly modify the state of other objects is called loosely coupled. Loose coupling makes the code more flexible, changeable, and easier to work with.

Objects that depend on other objects and can modify the states of other objects are called tightly coupled. It creates conditions where modifying the code of one object also requires changing the code of other objects. The reuse of code is difficult in tight coupling because we cannot separate the code.

Since using loose coupling is always a good habit.

---

29) Name the operators that cannot be overload.

1. Scope Resolution Operator (::)
2. Ternary Operator (? :)
3. Member Access or Dot Operator (.)
4. Pointer to Member Operator (.*)
5. sizeof operator

---

30) What is the difference between new and override?

The new modifier instructs the compiler to use the new implementation instead of the base class function. Whereas, Override modifier helps to override the base class function.

virtual: indicates that a method may be overridden by an inheritor

override: Overrides the functionality of a virtual method in a base class, providing different functionality.

new: Hides the original method (which doesn't have to be virtual), providing different functionality. This should only be used where it is absolutely necessary.

When you hide a method, you can still access the original method by upcasting to the base class. This is useful in some scenarios, but dangerous.

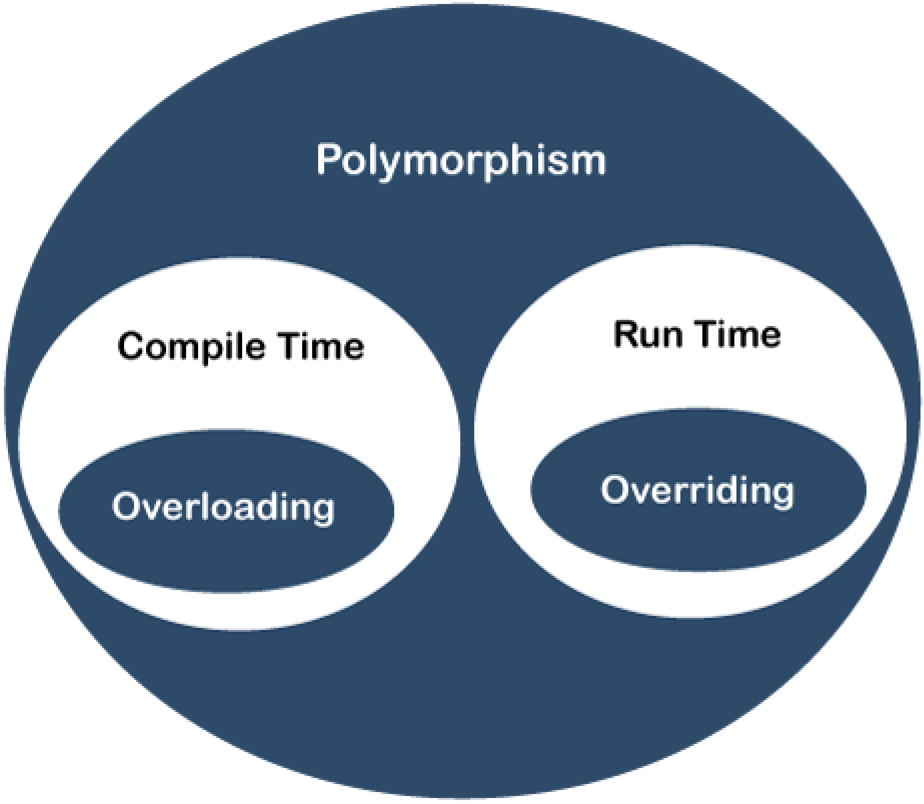
---

31) Explain overloading and overriding with example?

Overloading

Overloading is a concept in OOP when two or more methods in a class with the same name but the method signature is different. It is also known as compile-time polymorphism. For example, in the following code snippet, the method add() is an overloaded method.

```
public class Sum
{
int a, b, c;
public int add();
{
c=a+b;
return c;
}
add(int a, int b);
{
//logic
}
add(int a, int b, int c);
{
//logic
}
add(double a, double b, double c);
{
//logic
}
//statements
}
```



Overriding

If a method with the same method signature is presented in both child and parent class is known as method overriding. The methods must have the same number of parameters and the same type of parameter. It overrides the value of the parent class method. It is also known as runtime polymorphism. For example, consider the following program.

```
class Dog
{
public void bark()
{
System.out.println("woof ");
}
}
class Hound extends Dog
{
public void sniff()
{
System.out.println("sniff ");
```

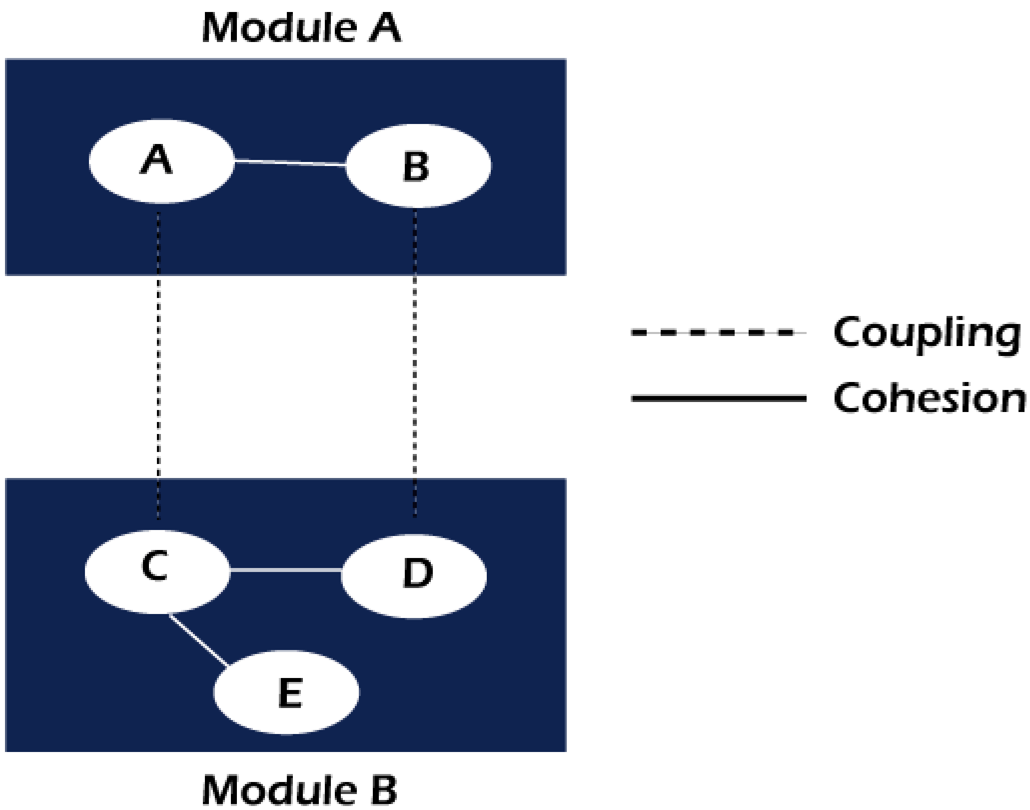
```java
}
//overrides the method bark() of the Dog class
public void bark()
{
System.out.println("bowl");
}
}
public class OverridingExample
{
public static void main(String args[])
{
Dog dog = new Hound();
//invokes the bark() method of the Hound class
dog.bark();
}
}
```

---

32) What is Cohesion in OOP?

In OOP, cohesion refers to the degree to which the elements inside a module belong together. It measures the strength of the relationship between the module and data. In short, cohesion represents the clarity of the responsibilities of a module. It is often contrasted with coupling.

It focuses on a how single module or class is intended. Higher the cohesiveness of the module or class, better is the object-oriented design.



There are two types of cohesion, i.e. High and Low.

- High cohesion is associated with several required qualities of software including robustness, reliability, and understandability.
- Low cohesion is associated with unwanted qualities such as being difficult to maintain, test, reuse, or even understand.

High cohesion often associates with loose coupling and vice versa.

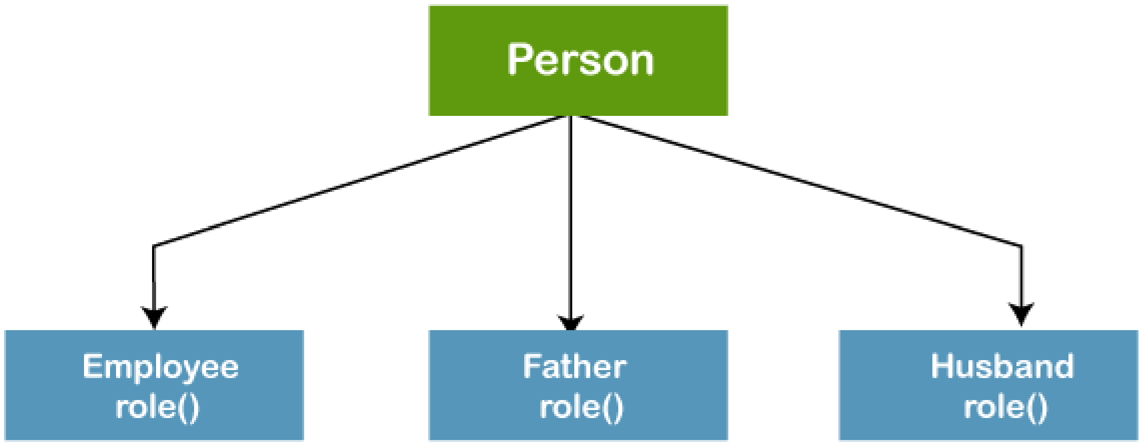
---

33) Give a real-world example of polymorphism?

The general meaning of Polymorphism is one that has different forms. The best real-world example of polymorphism is a person that plays different roles at different palaces or situations.

- At home a person can play the role of father, husband, and son.
- At the office the same person plays the role of boss or employee.
- In public transport, he plays the role of passenger.
- In the hospital, he can play the role of doctor or patient.

- At the shop, he plays the role of customer.



Hence, the same person possesses different behavior in different situations. It is called polymorphism.

---

34) What is the difference between a base class and a superclass?

The base class is the root class- the most generalized class. At the same time, the superclass is the immediate parent class from which the other class inherits.

---

35) What is data abstraction and how can we achieve data abstraction?

It is one of the most important features of OOP. It allows us to show only essential data or information to the user and hides the implementation details from the user. A real-world example of abstraction is driving a car. When we drive a car, we do not need to know how the engine works (implementation) we only know how ECG works.

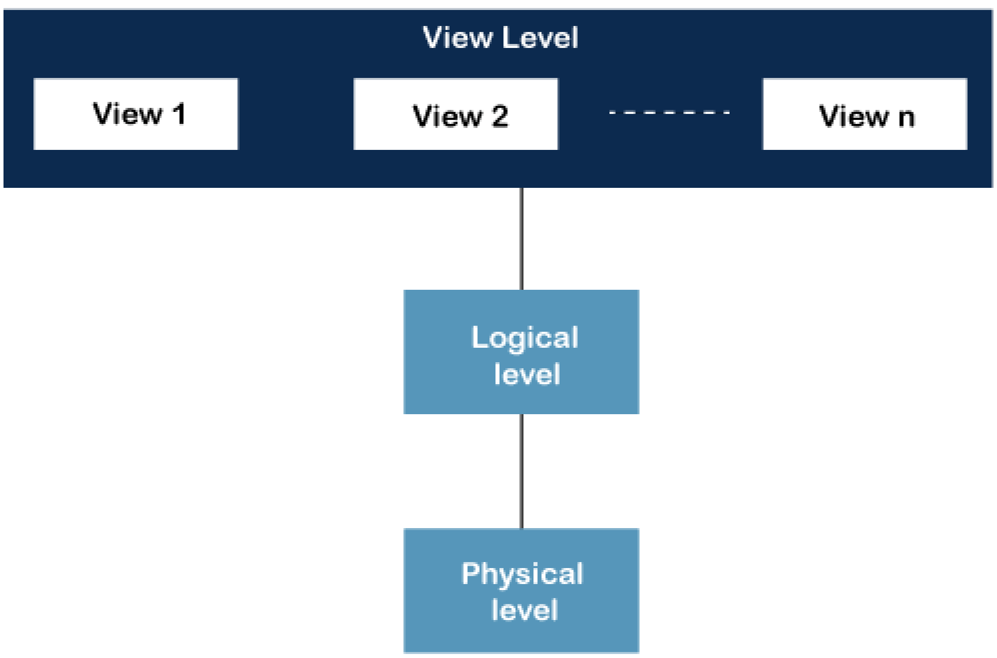
There are two ways to achieve data abstraction

- Abstract class
- Abstract method

36) What are the levels of data abstraction?

There are three levels of data abstraction:

- Physical Level: It is the lowest level of data abstraction. It shows how the data is actually stored in memory.
- Logical Level: It includes the information that is actually stored in the database in the form of tables. It also stores the relationship among the data entities in relatively simple structures. At this level, the information available to the user at the view level is unknown.
- View Level: It is the highest level of data abstraction. The actual database is visible to the user. It exists to ease the availability of the database by an individual user.



Levels of data abstraction

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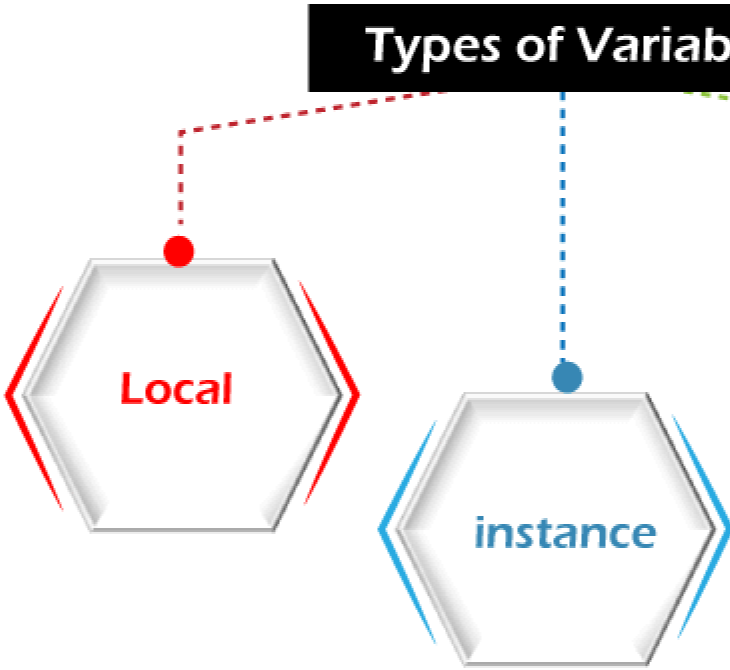
37) What are the types of variables in OOP?

There are three types of variables:



Instance Variable: It is an object-level variable. It should be declared inside a class but must be outside a method, block, and constructor. It is created when an object is created by using the new keyword. It can be accessed directly by calling the variable name inside the class.

Static Variable: It is a class-level variable. It is declared with keyword static inside a class but must be outside of the method, block, and constructor. It stores in static memory. Its visibility is the same as the instance variable. The default value of a static variable is the same as the instance variable. It can be accessed by calling the class_name.variable_name.

Local Variable: It is a method-level variable. It can be declared in method, constructor, or block. Note that the use of an access modifier is not allowed with local variables. It is visible only to the method, block, and constructor in which it is declared. Internally, it is implemented at the stack level. It must be declared and initialized before use.

Another type of variable is used in object-oriented programming is the reference variable.

Reference Variable: It is a variable that points to an object of the class. It points to the location of the object that is stored in the memory.

38) Is it possible to overload a constructor?

Yes, the constructors can be overloaded by changing the number of arguments accepted by the constructor or by changing the data type of the parameters. For example:

```
public class Demo
{
Demo()
{
//logic
}
Demo(String str) //overloaded constructor
{
//logic
}
Demo(double d) //overloaded constructor
{
//logic
}
//statements
}
```

---

39) Can we overload the main() method in Java also give an example?

Yes, we can also overload the main() method in Java. Any number of main() methods can be defined in the class, but the method signature must be different. Consider the following code.

```
class OverloadMain
{
public static void main(int a)  //overloaded main method
{
System.out.println(a);
}
public static void main(String args[])
{
System.out.println("main method invoked");
```

```
main(6);
}
}
```

---

40) Consider the following scenario:

If a class Demo has a static block and a main() method. A print statement is presented in both. The question is which one will first execute, static block or the main() method, and why?

JVM first executes the static block on a priority basis. It means JVM first goes to static block even before it looks for the main() method in the program. After that main() method will be executed.

```
class Demo
{
static              //static block
{
System.out.println("Static block");
}
public static void main(String args[])  //static method
{
System.out.println("Static method");
}
}
```
